TFM - Stacked y-maps and μy-maps

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2026

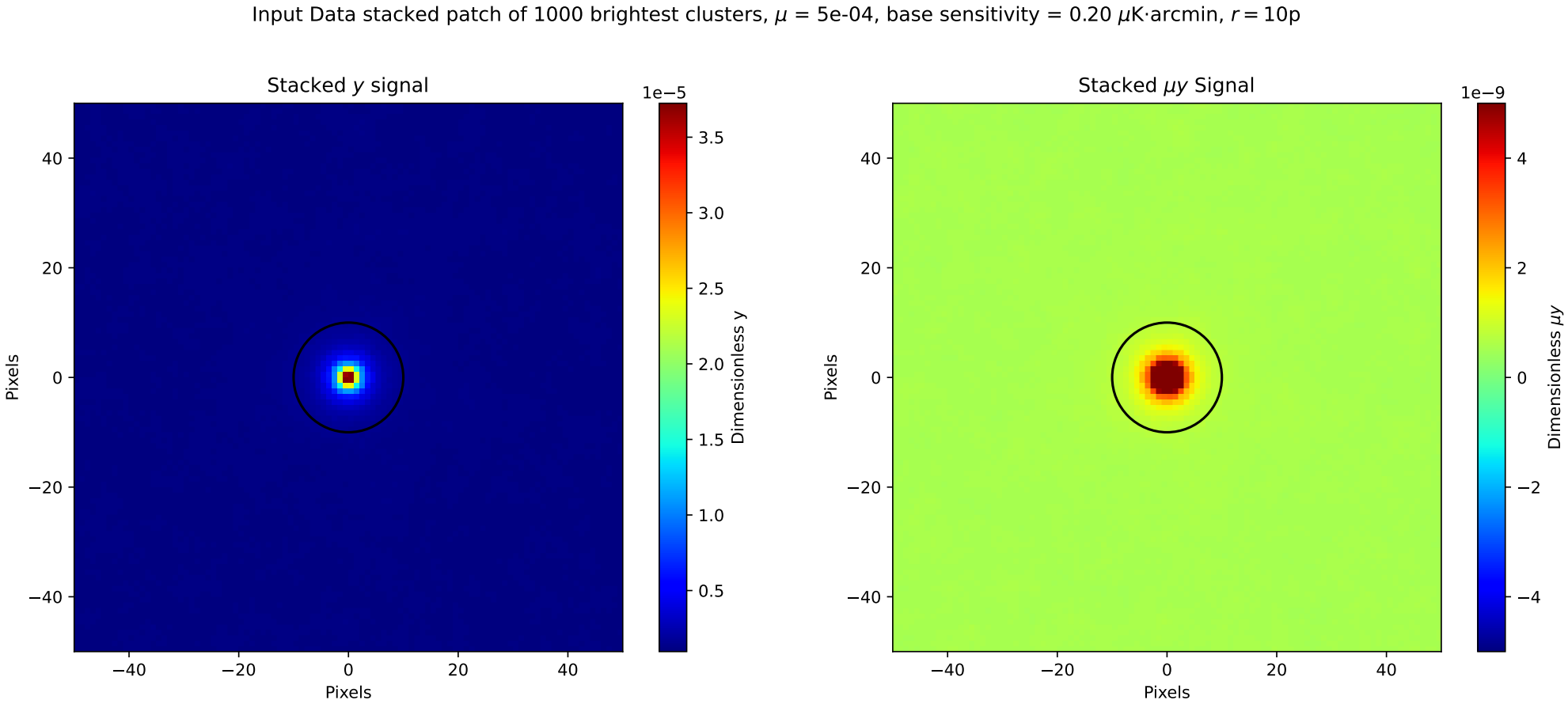
Universidad de Cantabria

Input stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 5 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3



ILC stacked zoomed maps

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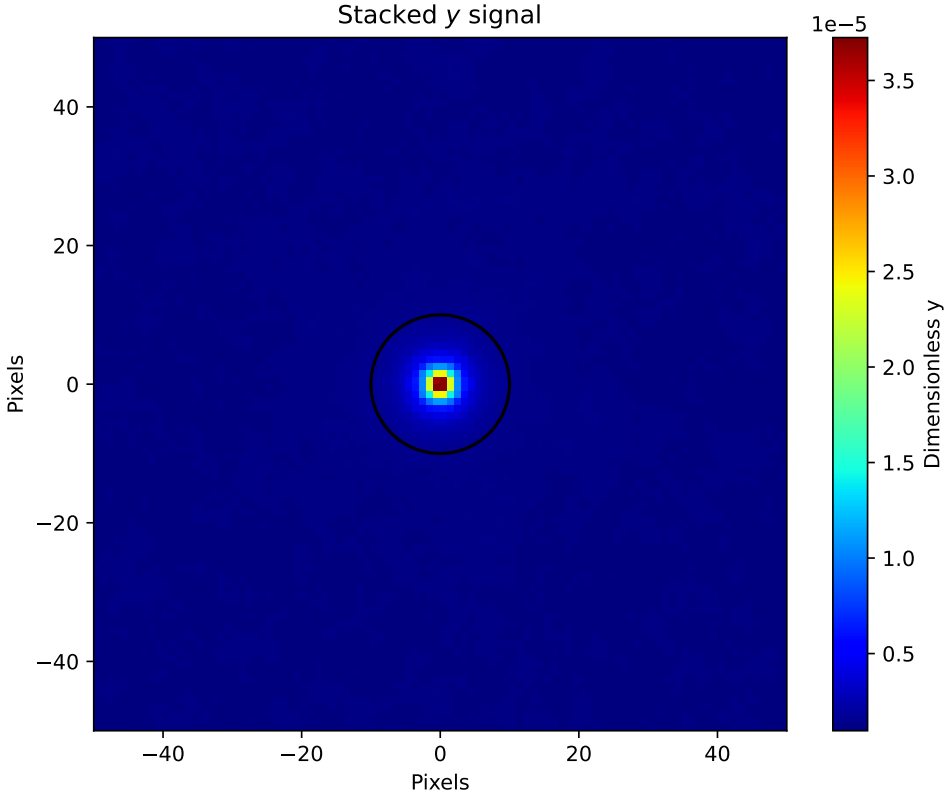
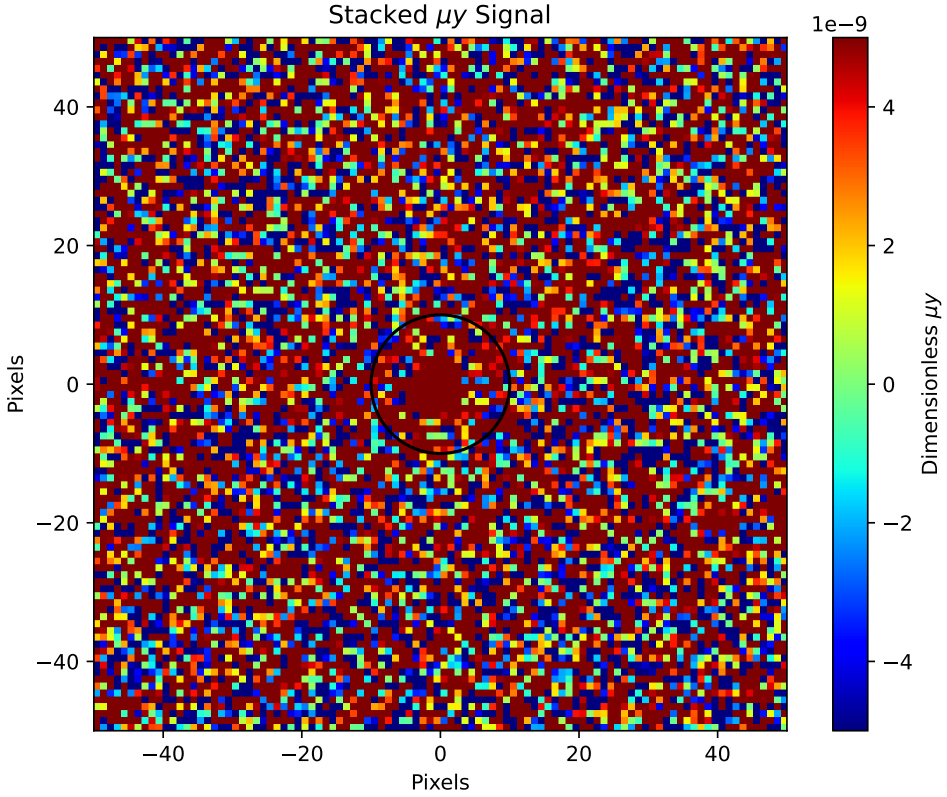
P-ILC stacked patch of 1000 brightest clusters, μ = 5e-04, base sensitivity = 0.20 μK·arcmin, r = 10p





CILC stacked zoomed maps

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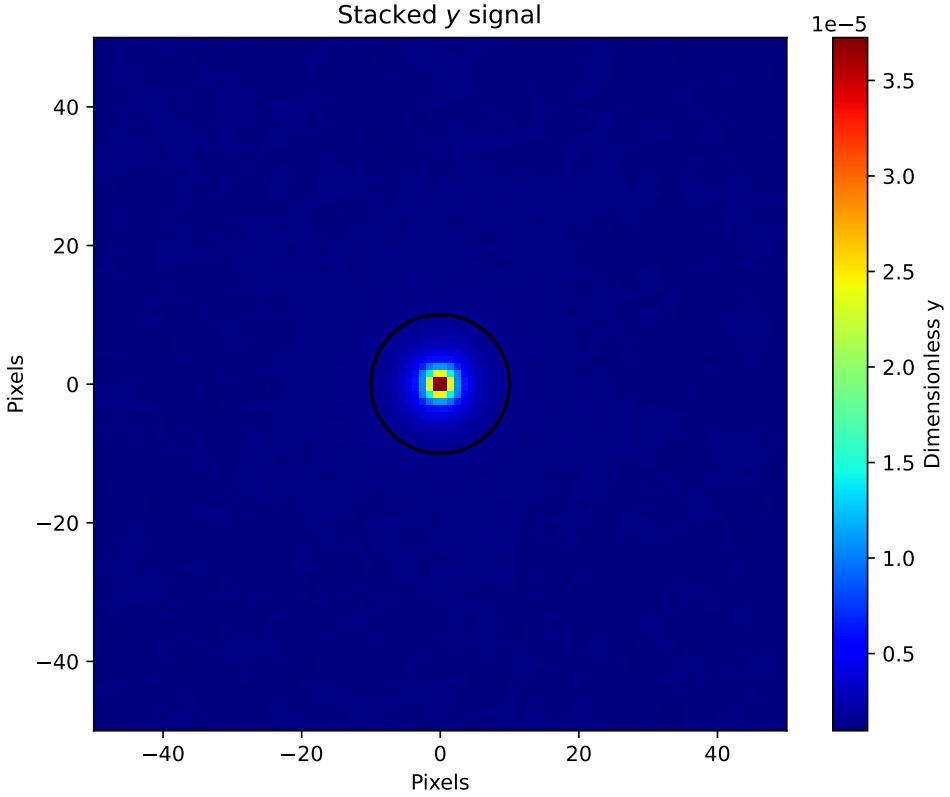

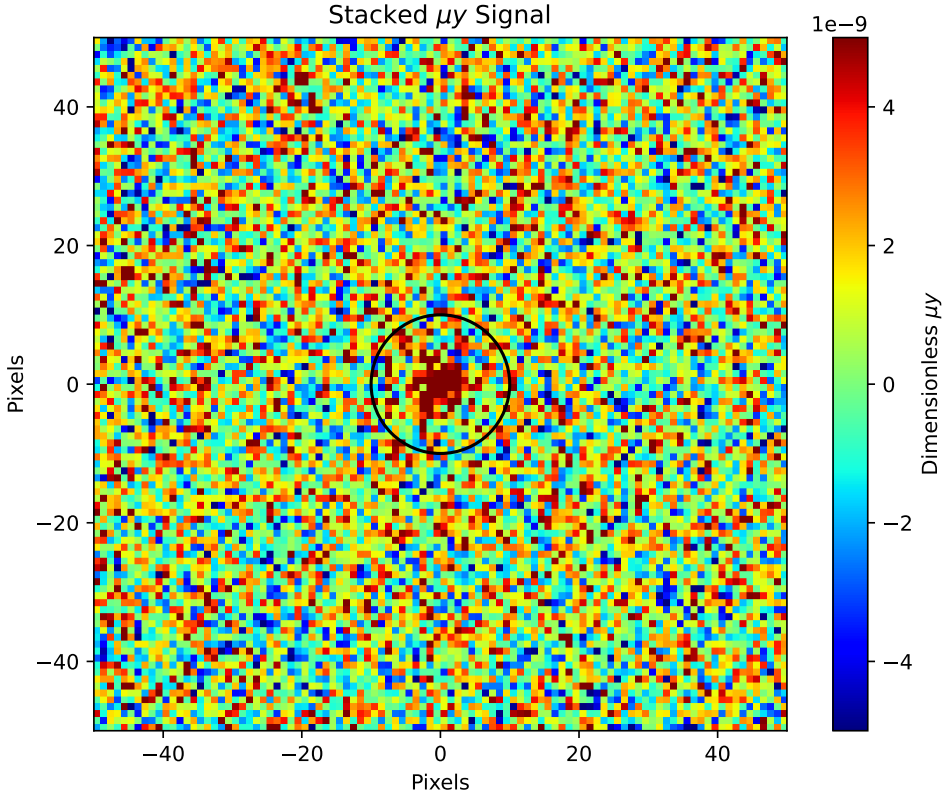

Comparison y stacked zoomed maps

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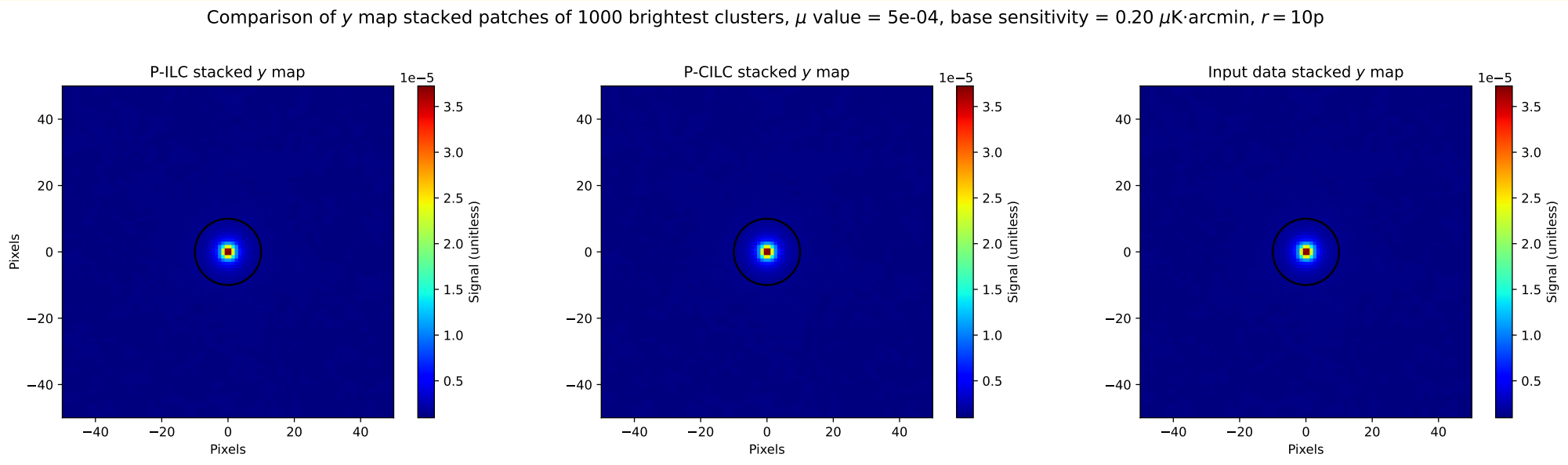

Comparison μy stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

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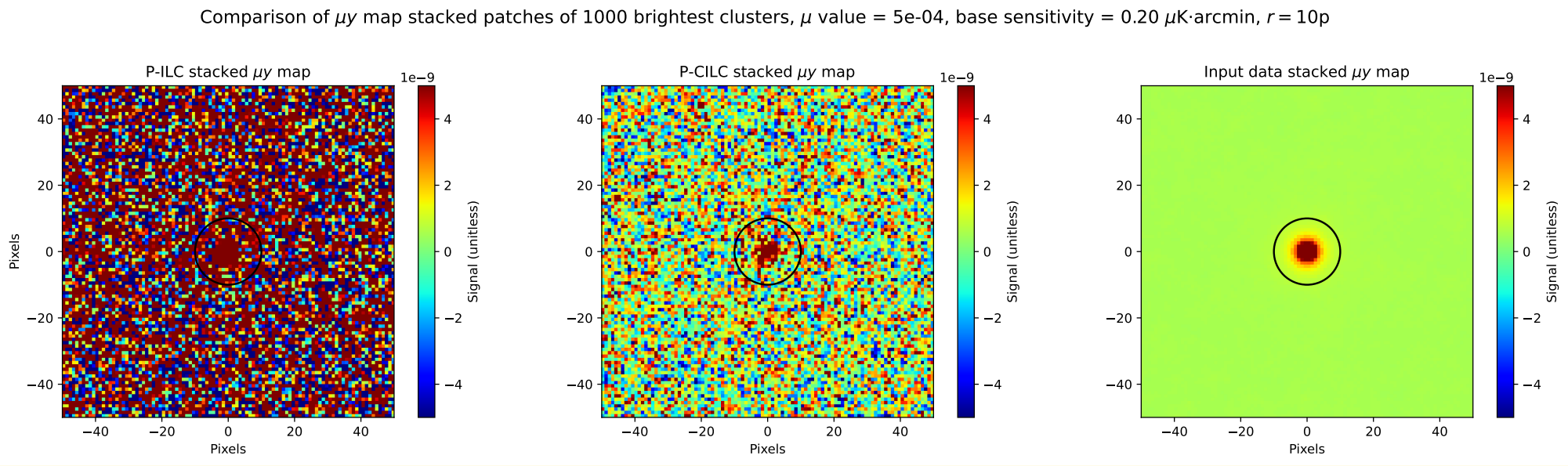

Input stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

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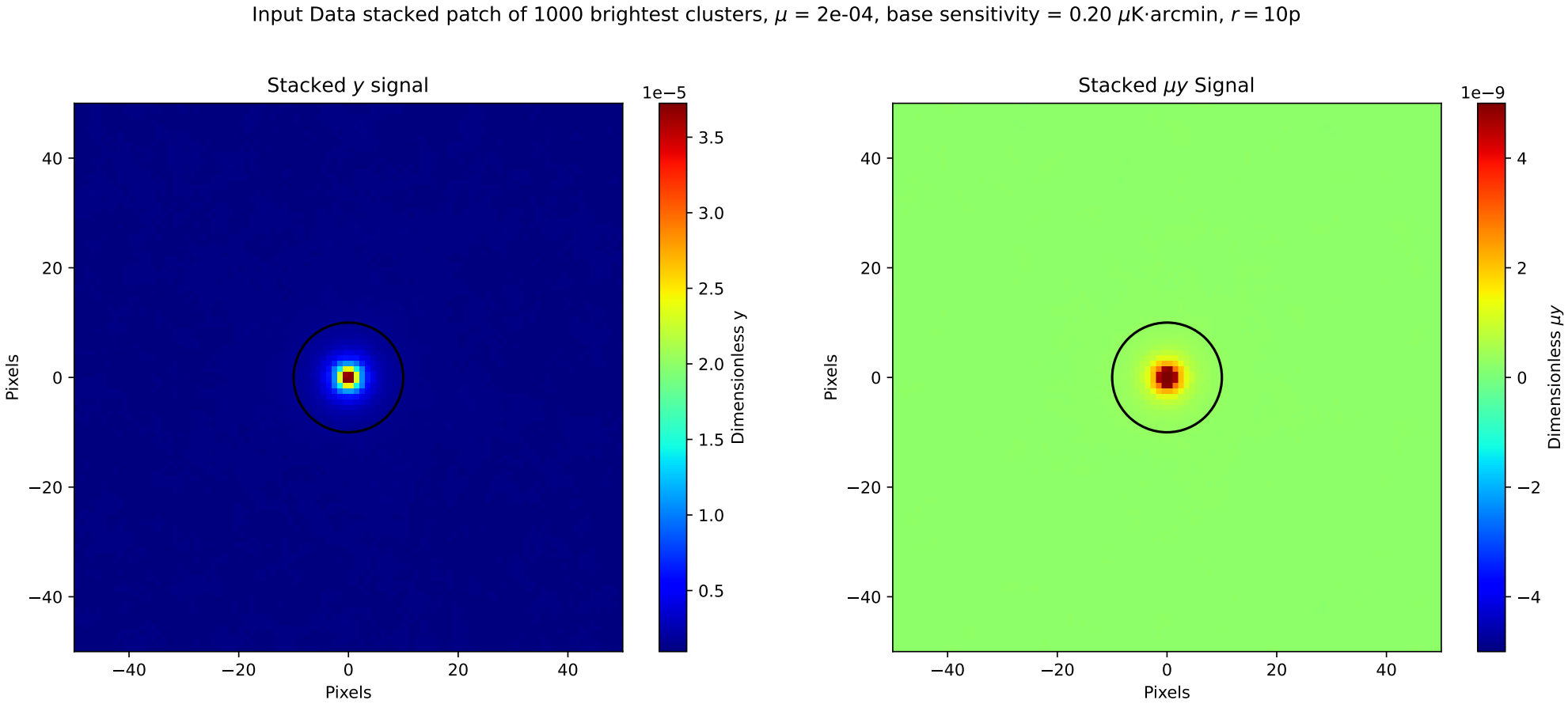

ILC stacked zoomed maps

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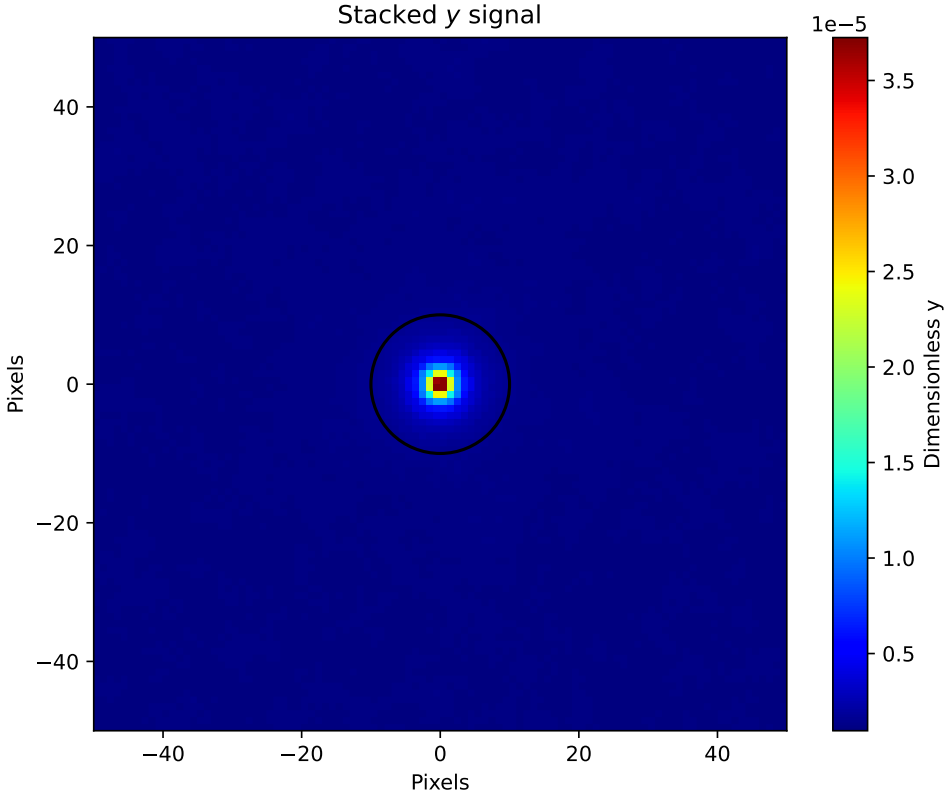
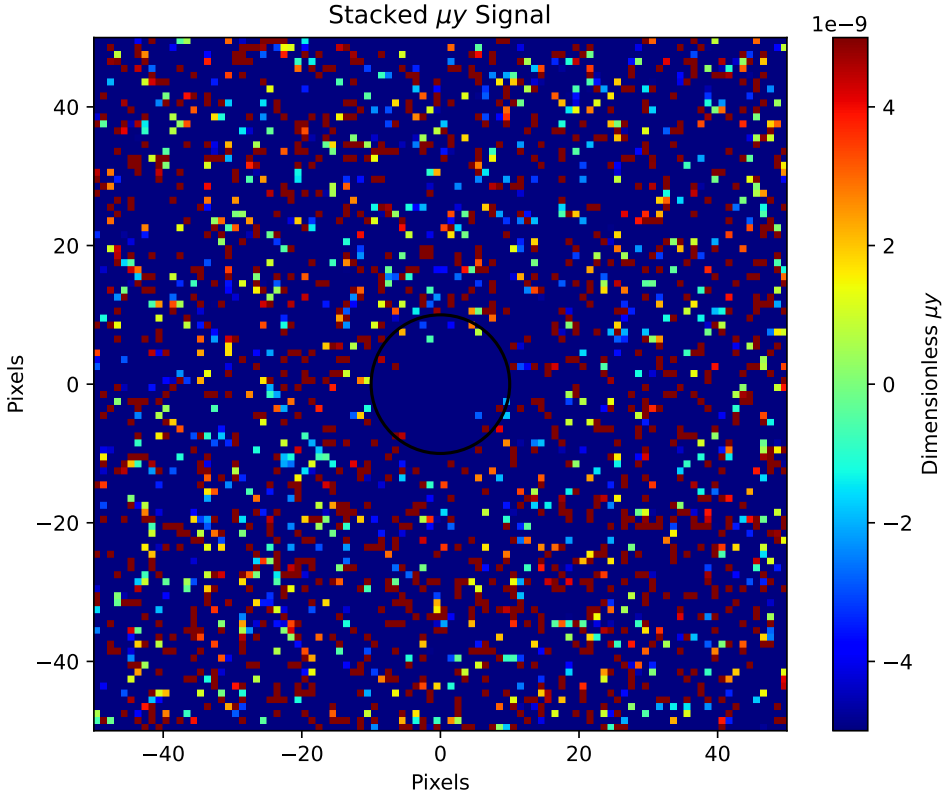
P-ILC stacked patch of 1000 brightest clusters, μ = 2e-04, base sensitivity = 0.20 μK·arcmin, r = 10p





CILC stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

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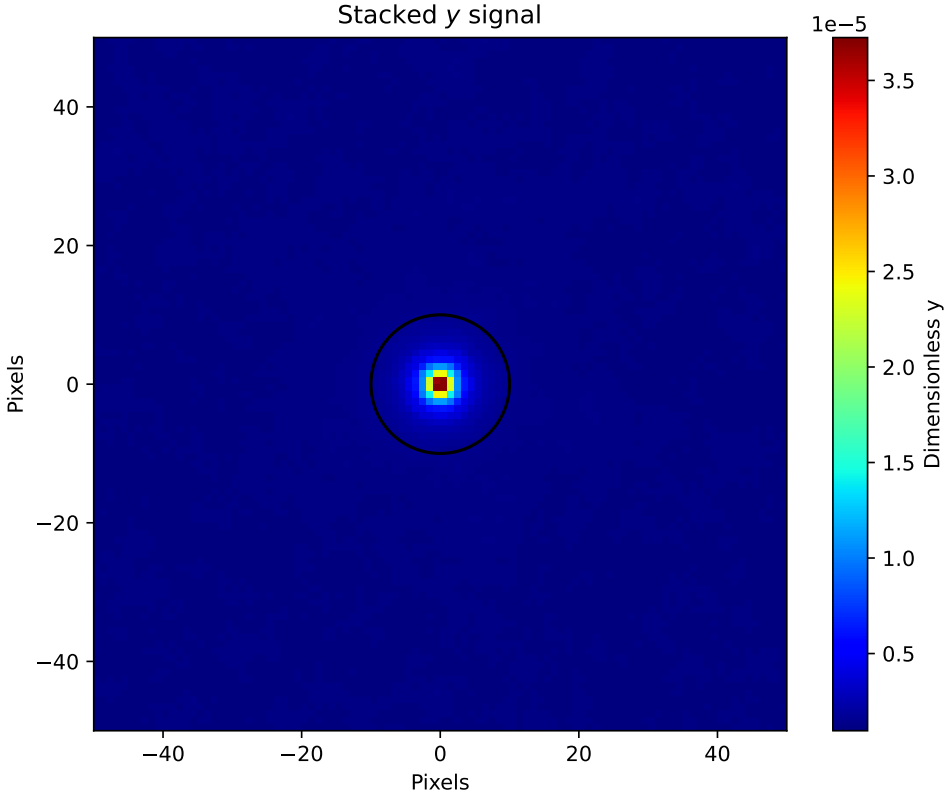

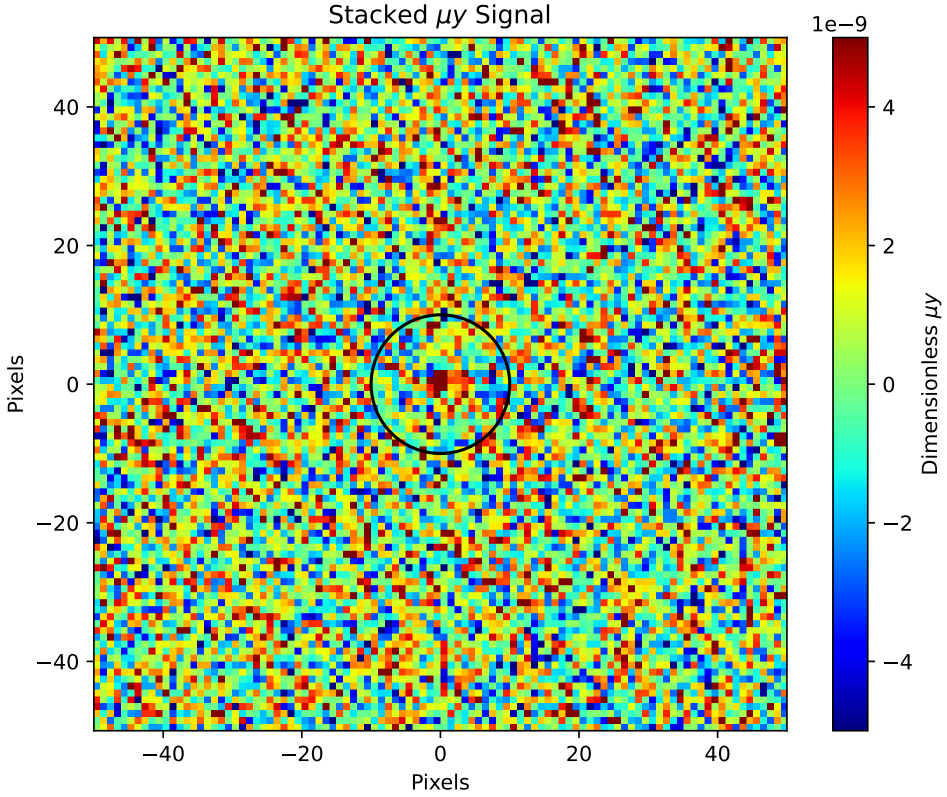

Comparison y stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

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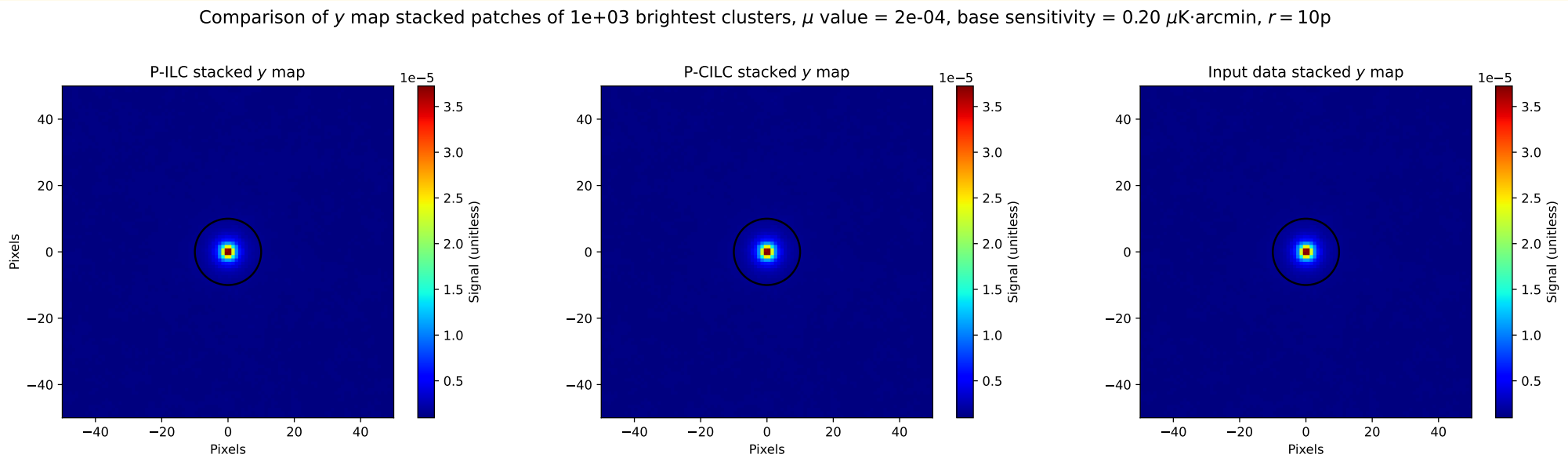

Comparison μy stacked zoomed maps

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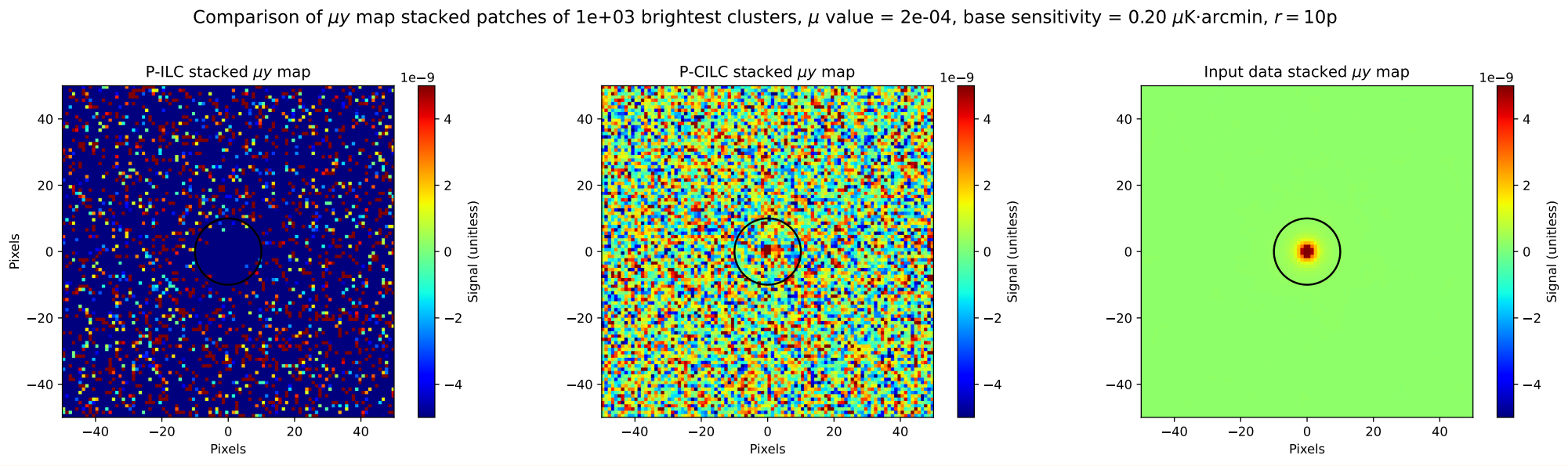

Input stacked zoomed maps

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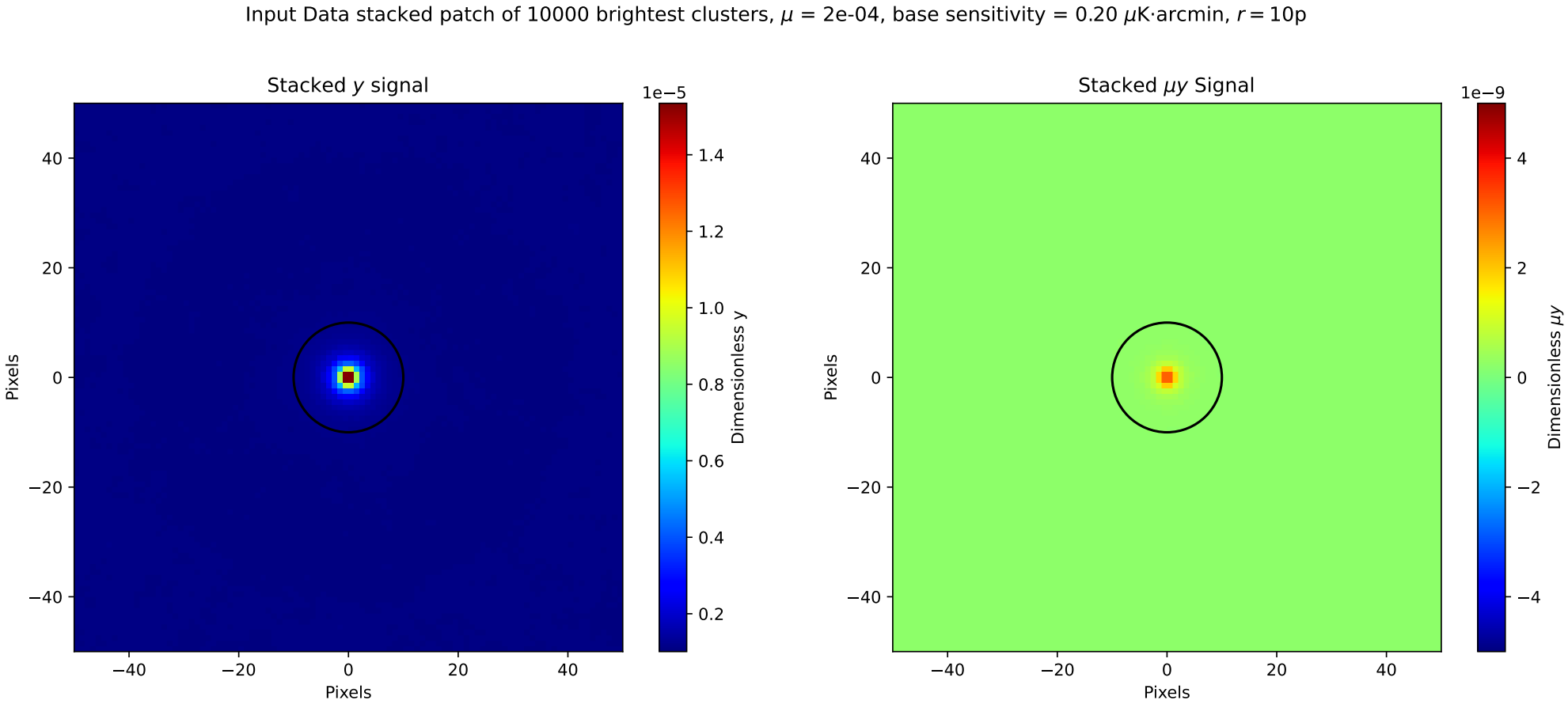
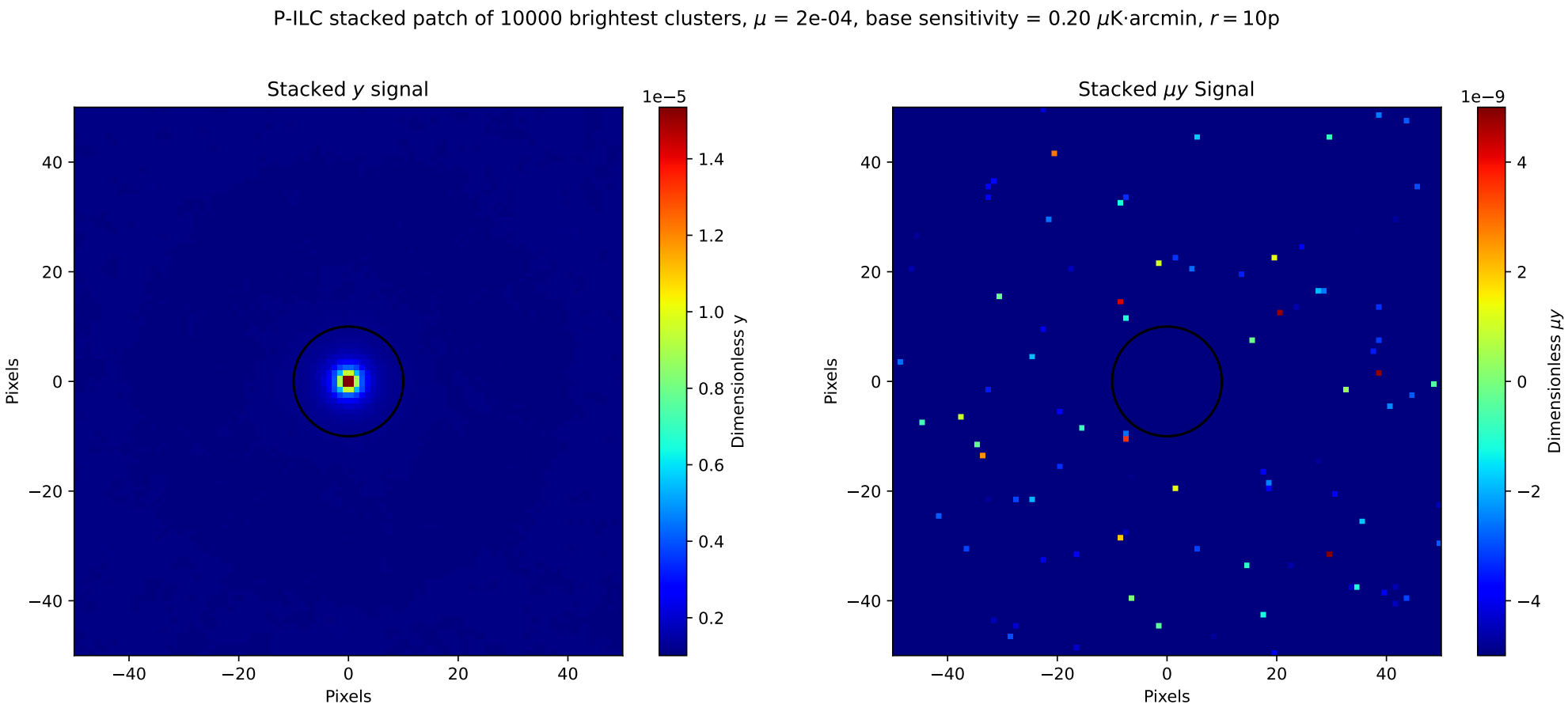


ILC stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4



CILC stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

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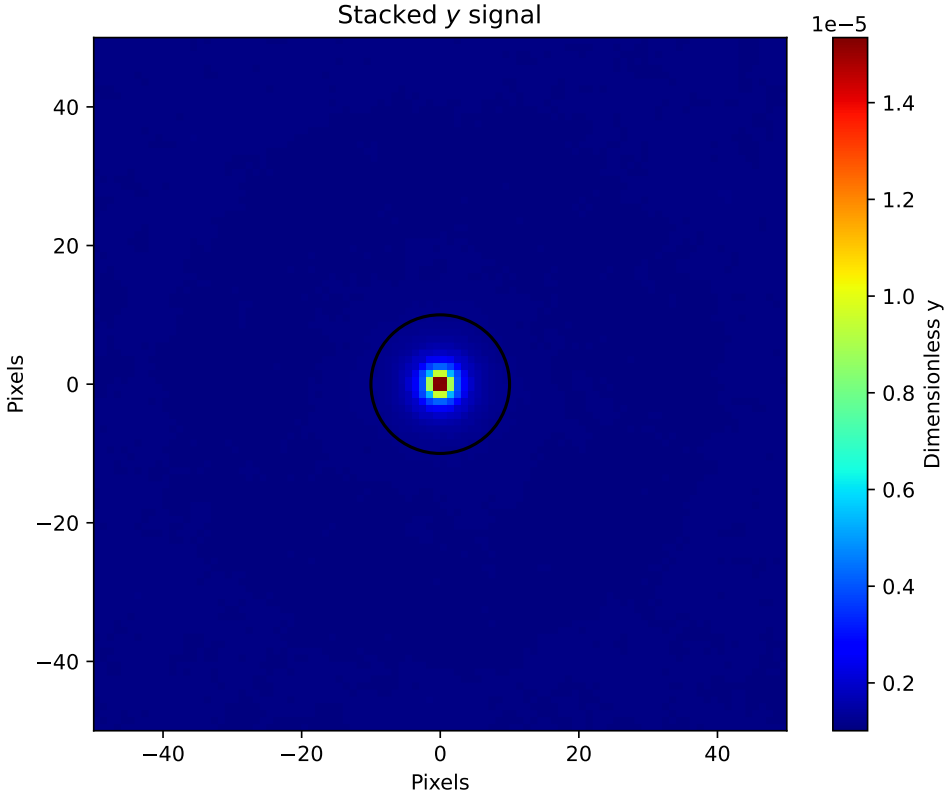
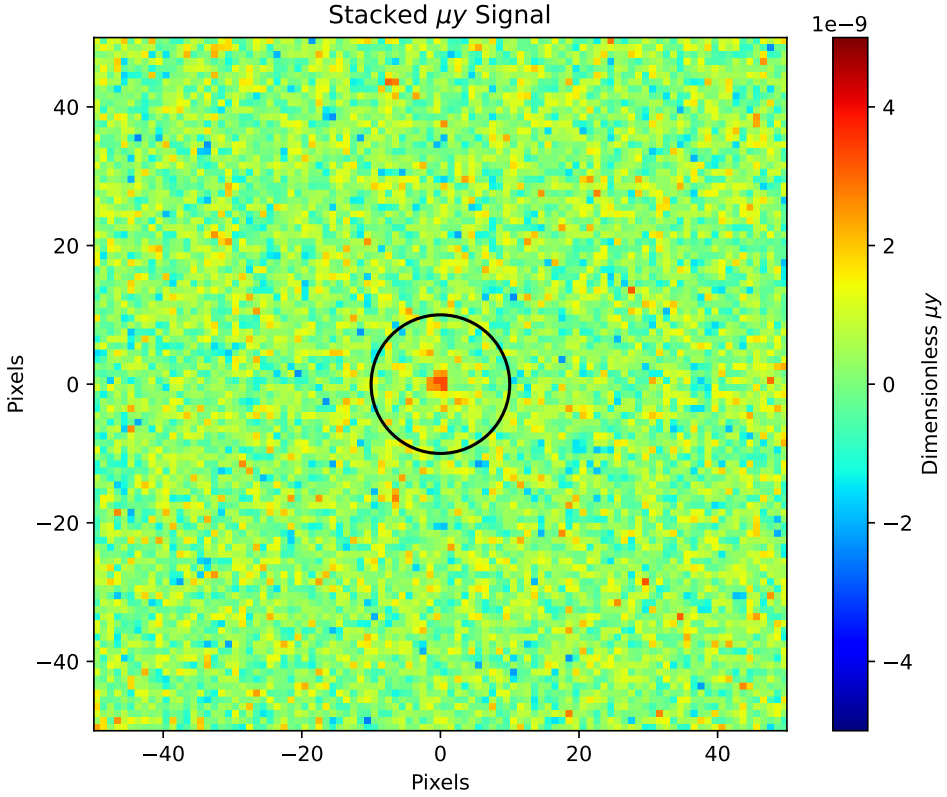
P-CILC stacked patch of 10000 brightest clusters, μ = 2e-04, base sensitivity = 0.20 μK·arcmin, r = 10p





Comparison y stacked zoomed maps

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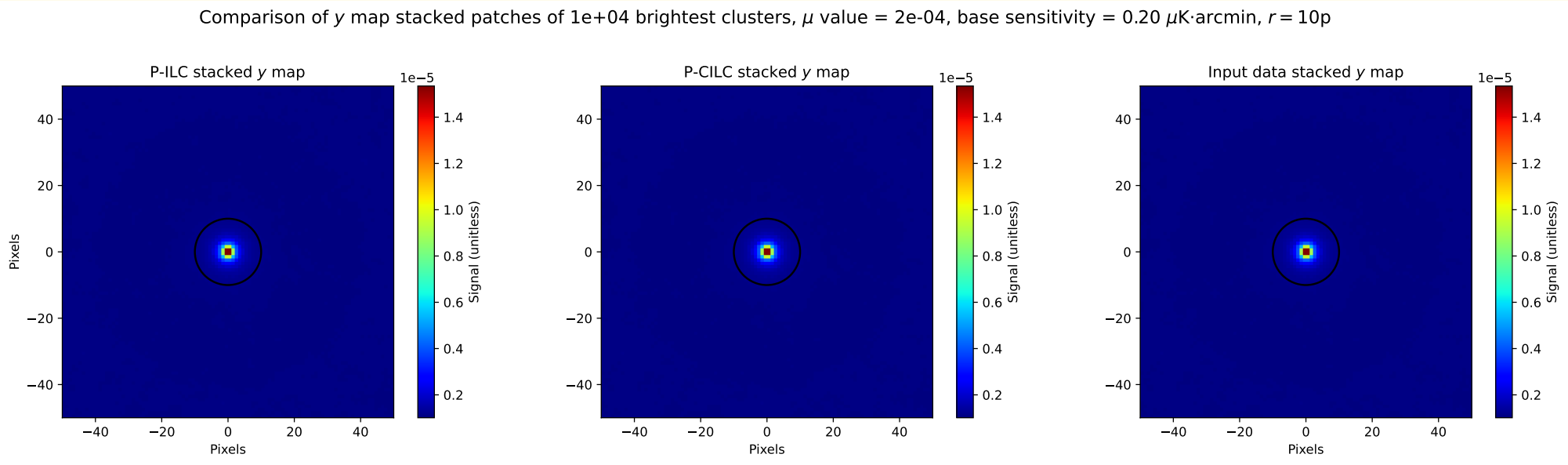

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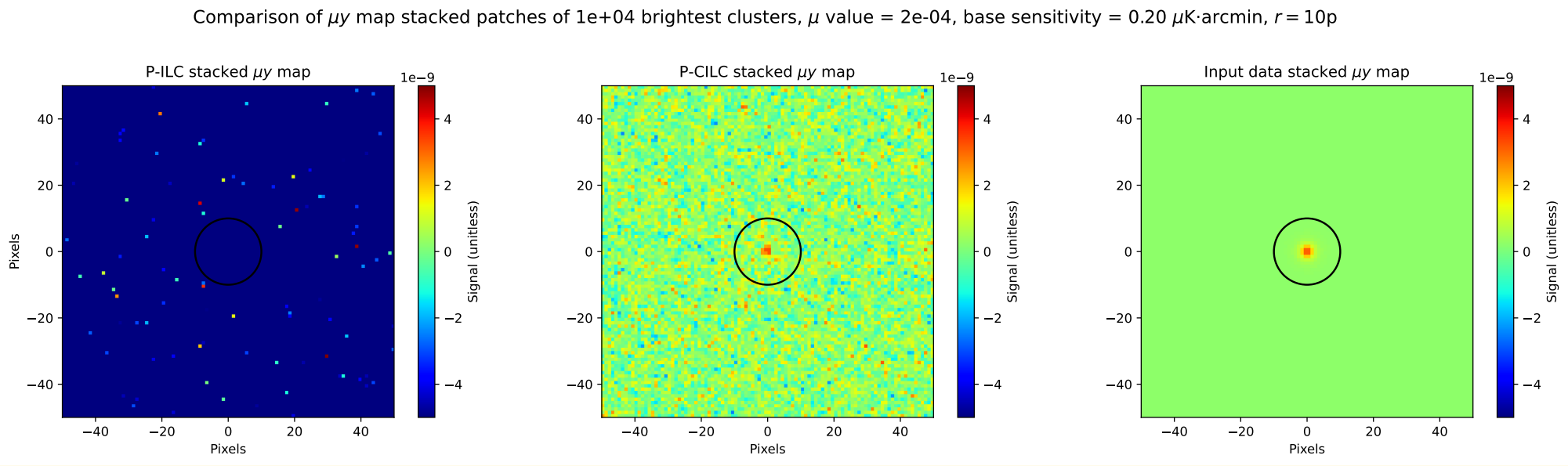
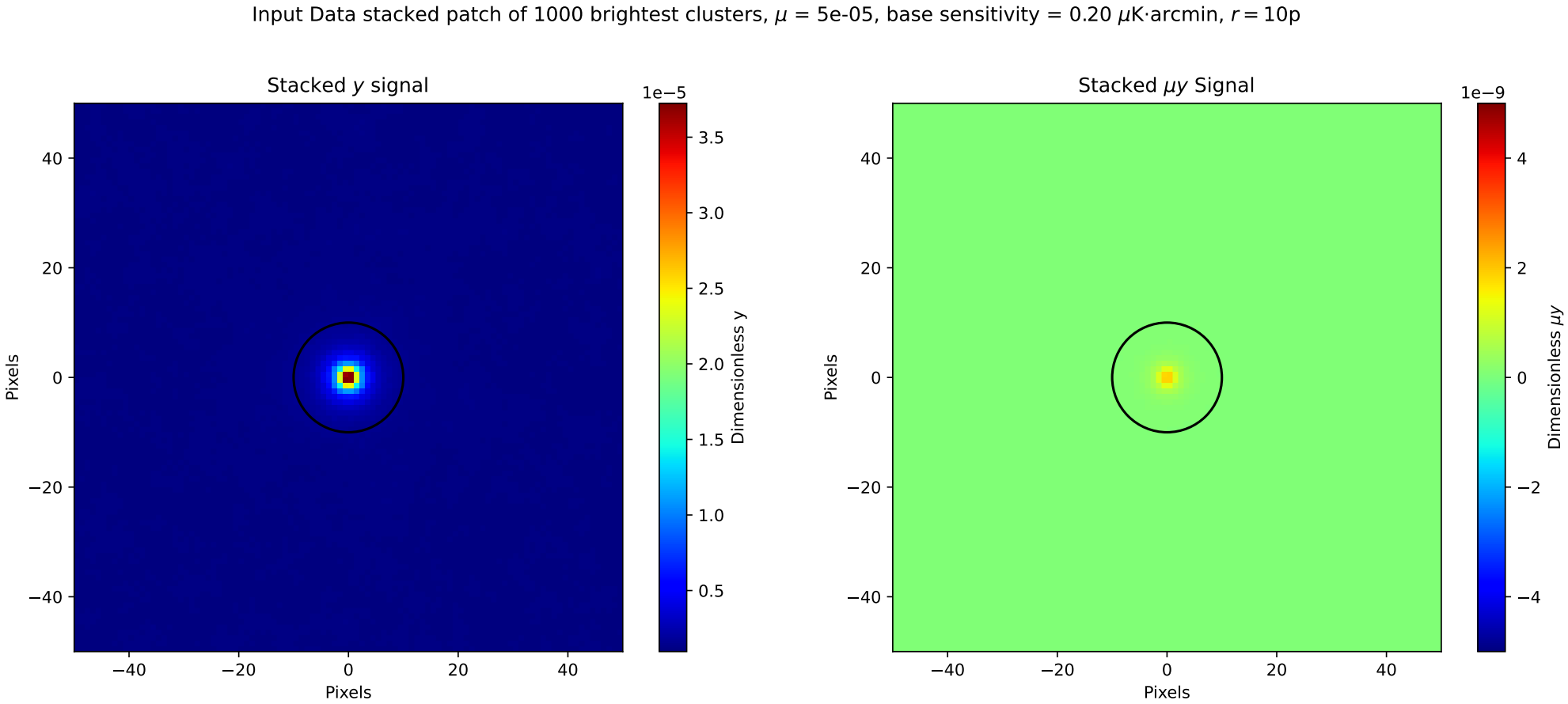


Input stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3



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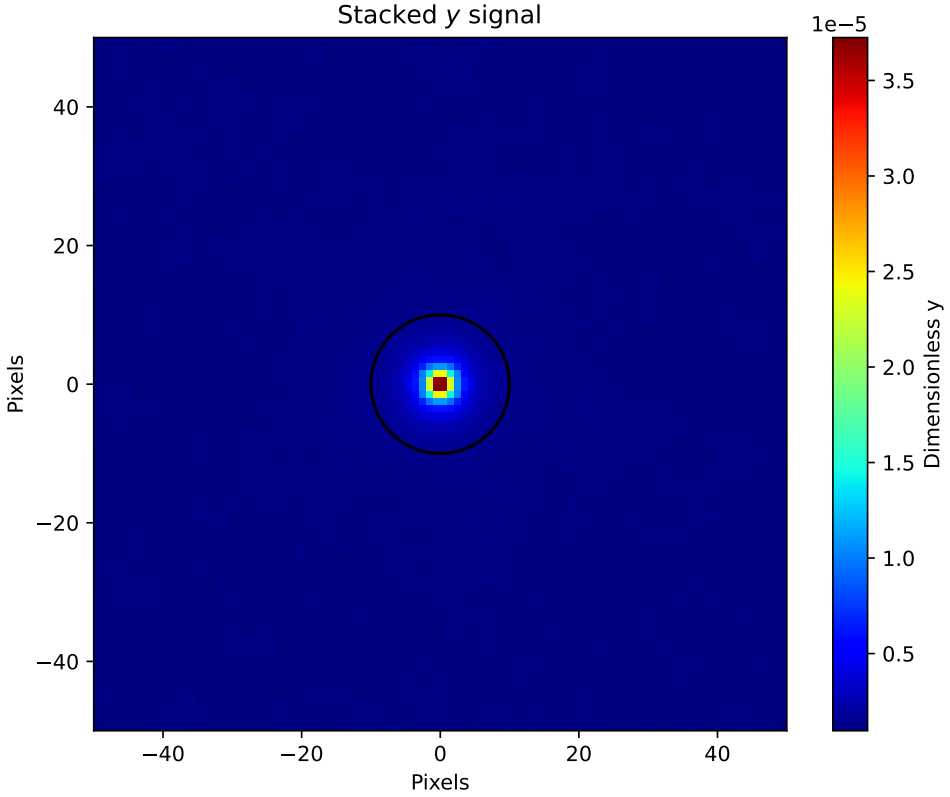

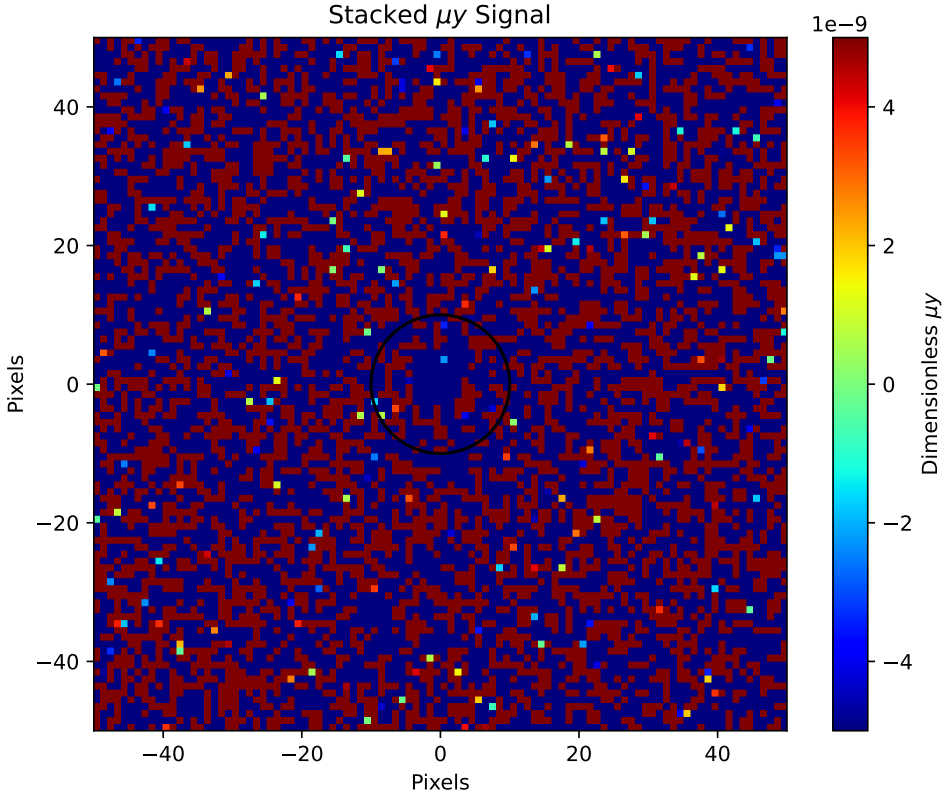

CILC stacked zoomed maps

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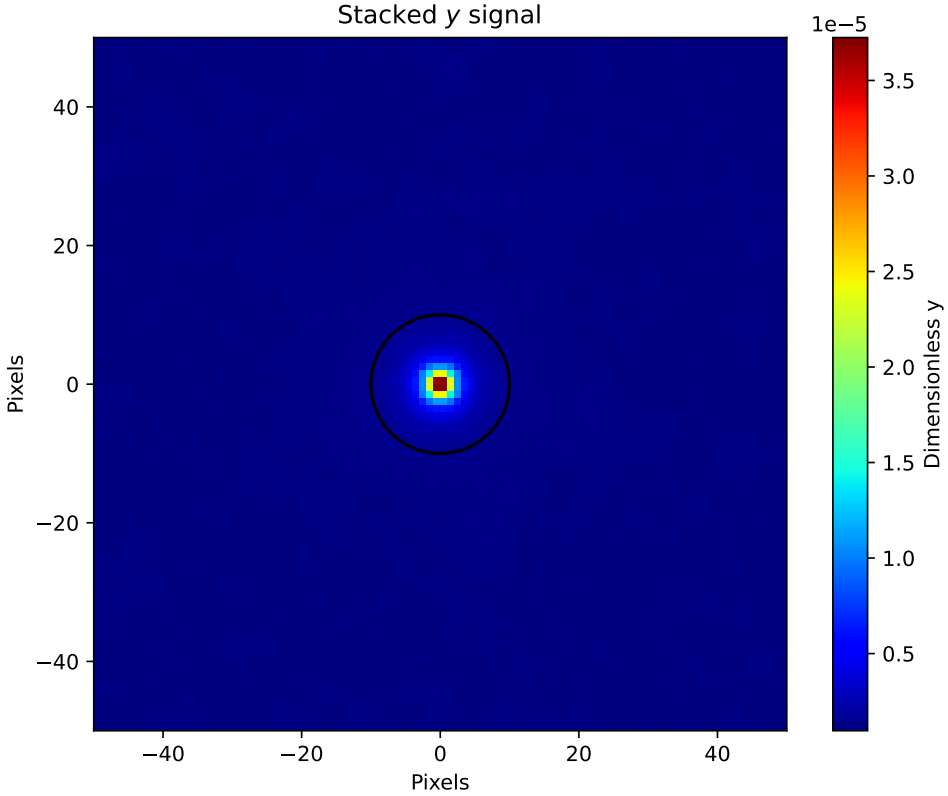

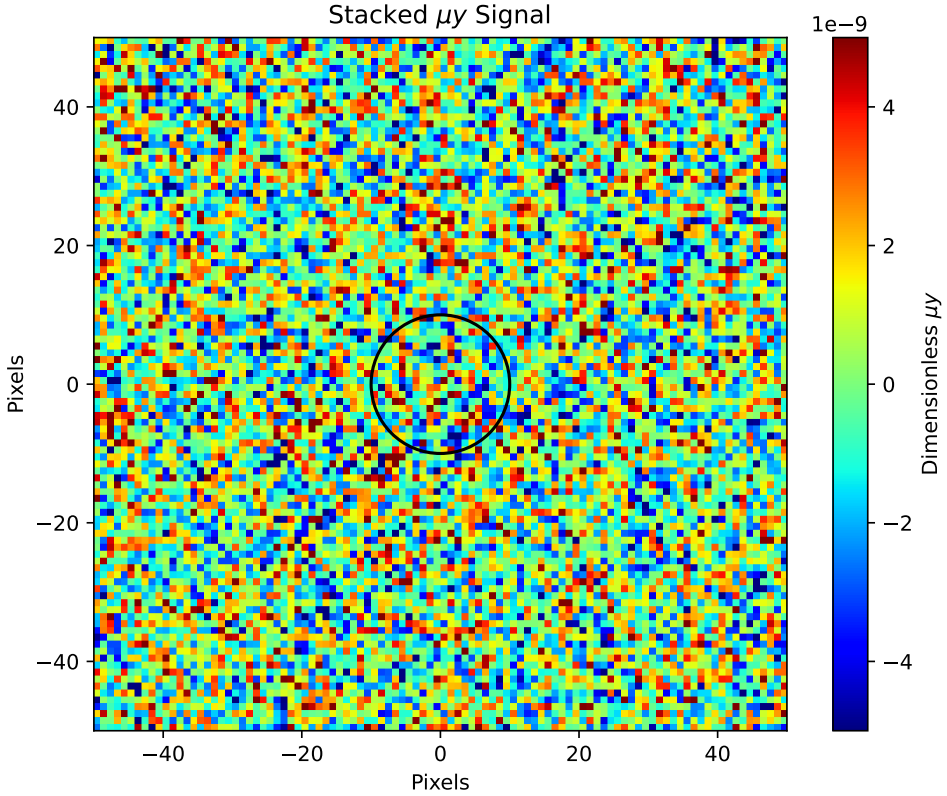

Comparison y stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

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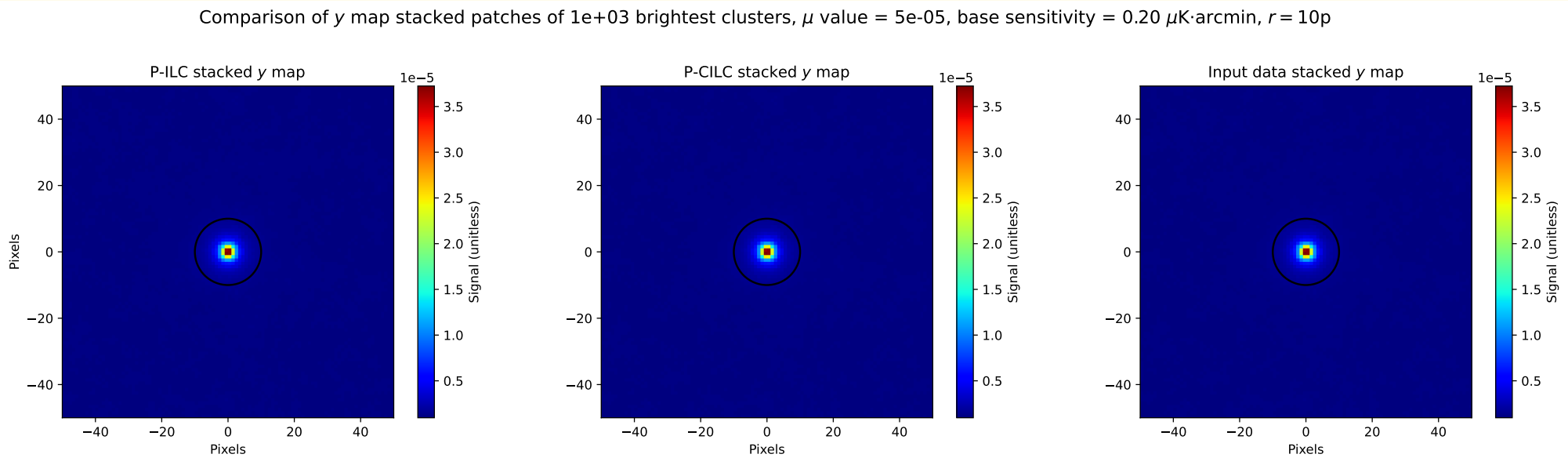

Comparison μy stacked zoomed maps

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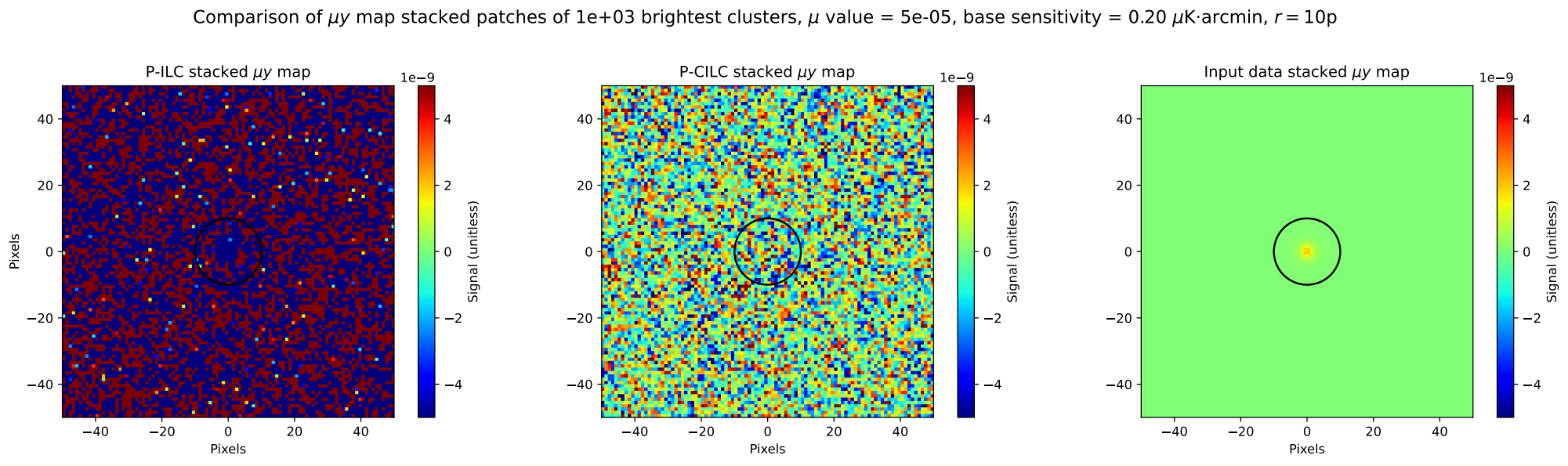

Input stacked zoomed maps

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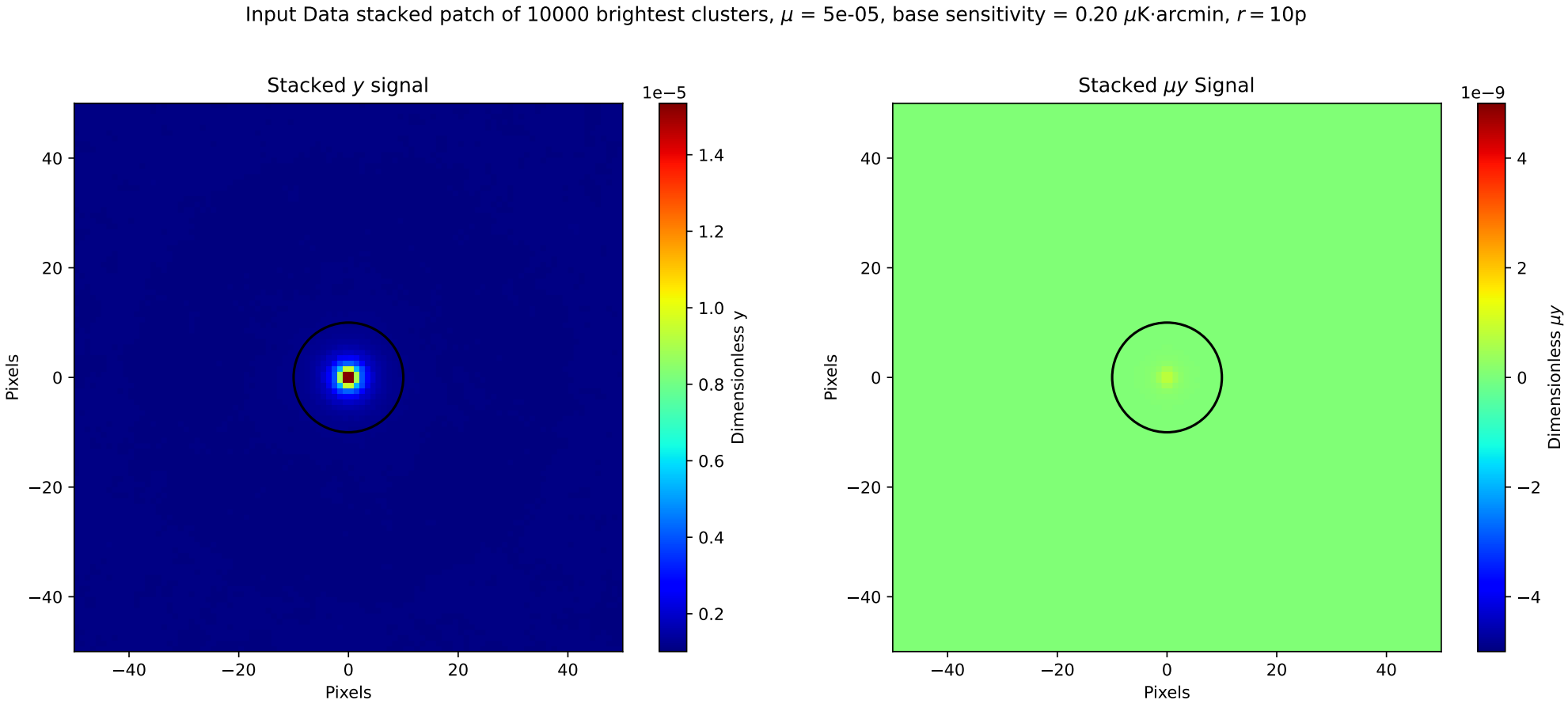

ILC stacked zoomed maps

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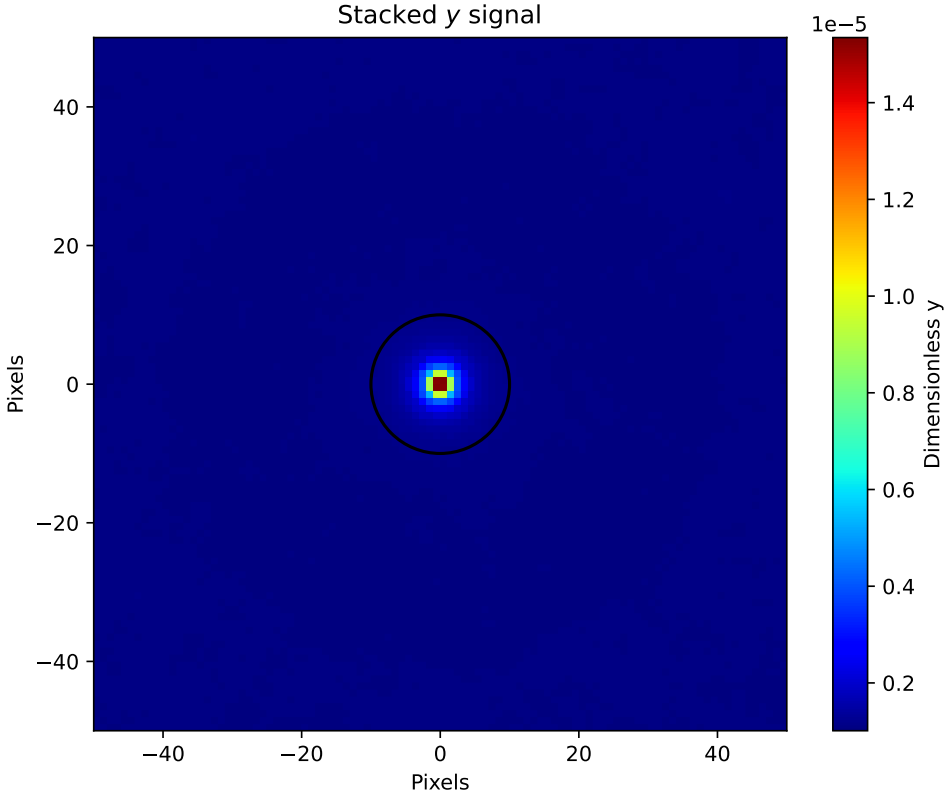

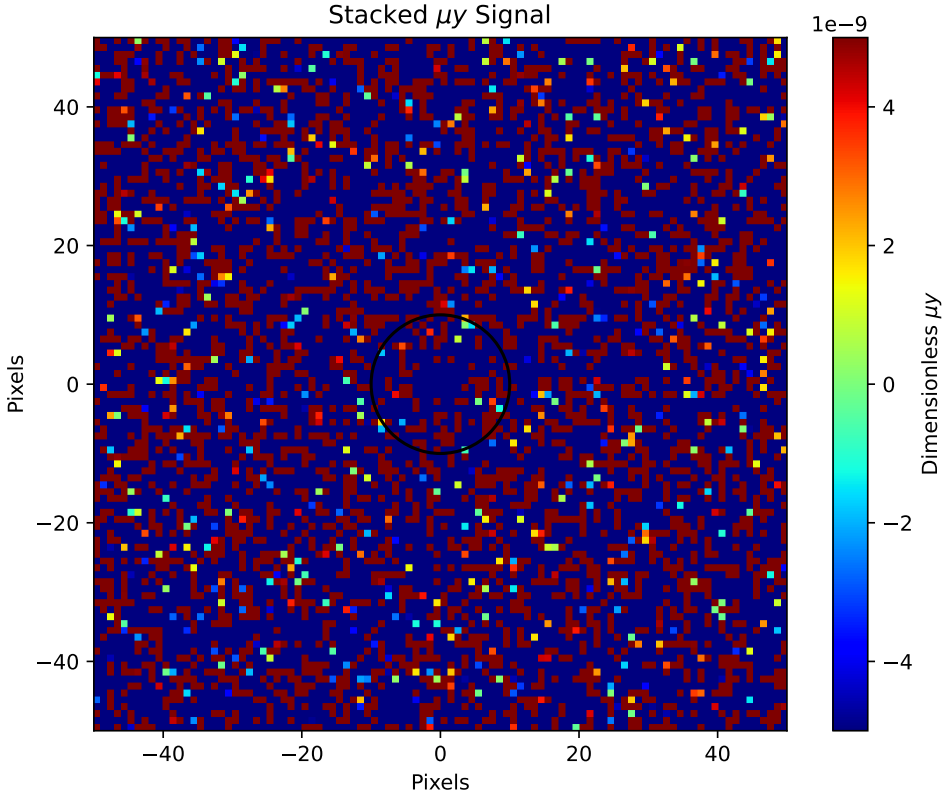

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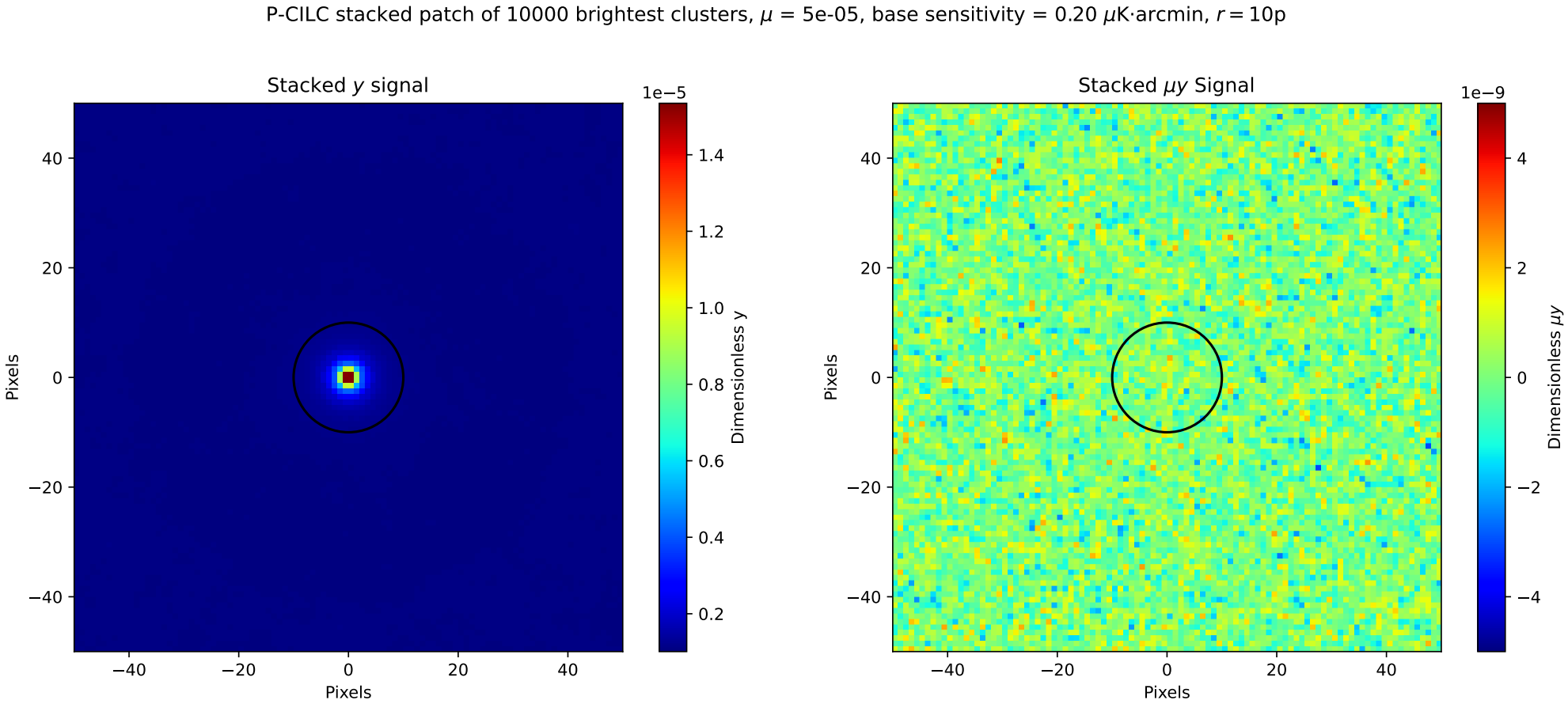

Comparison y stacked zoomed maps

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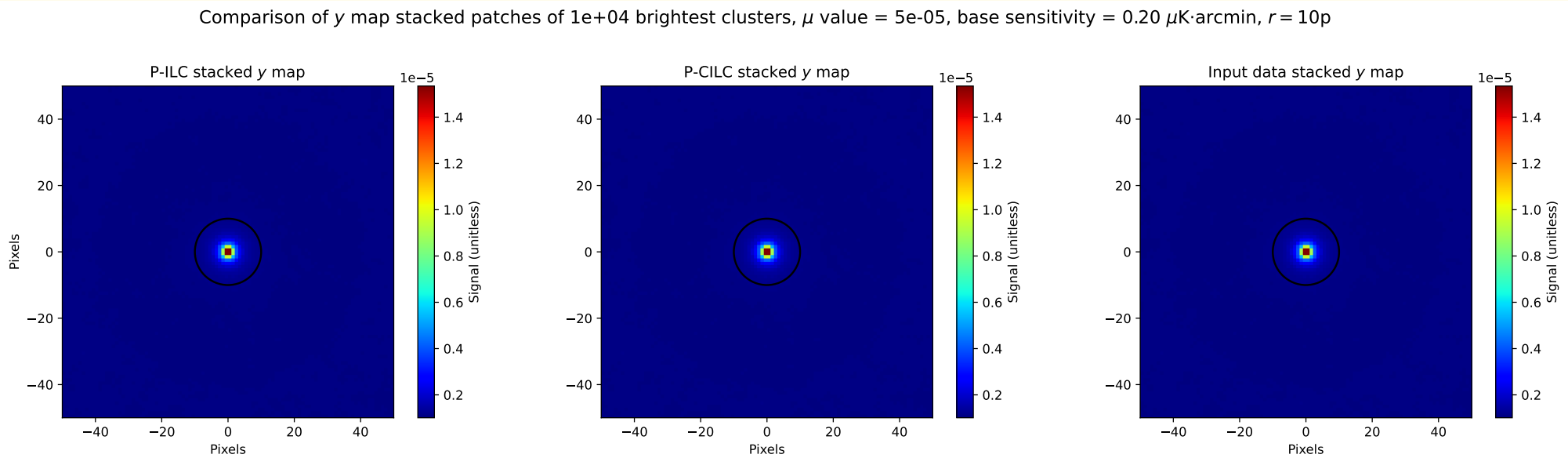

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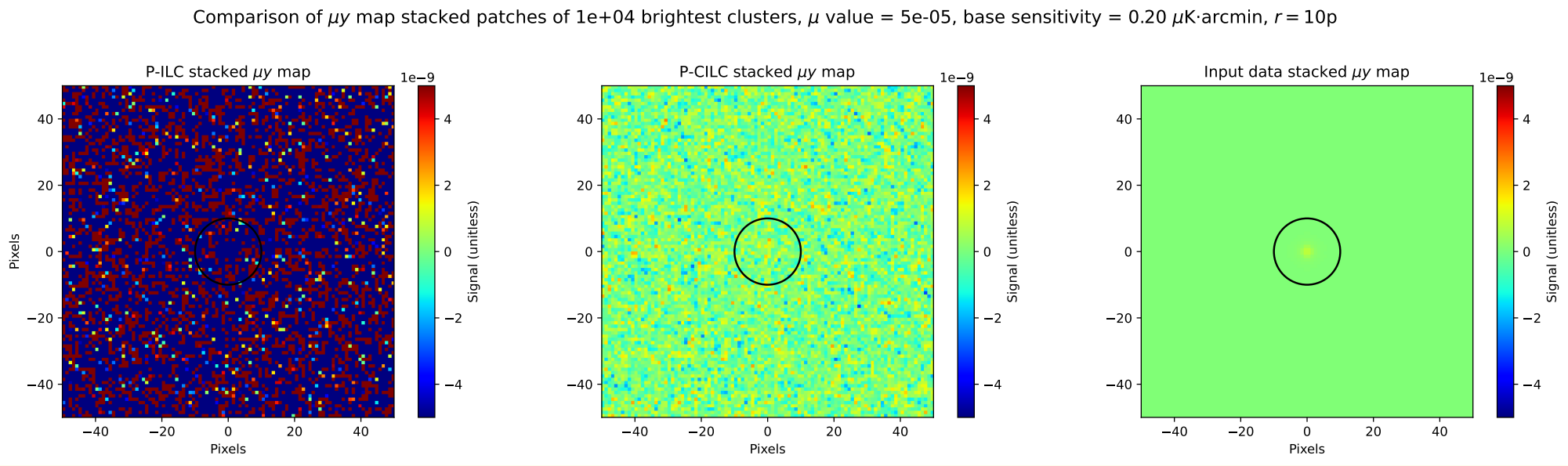
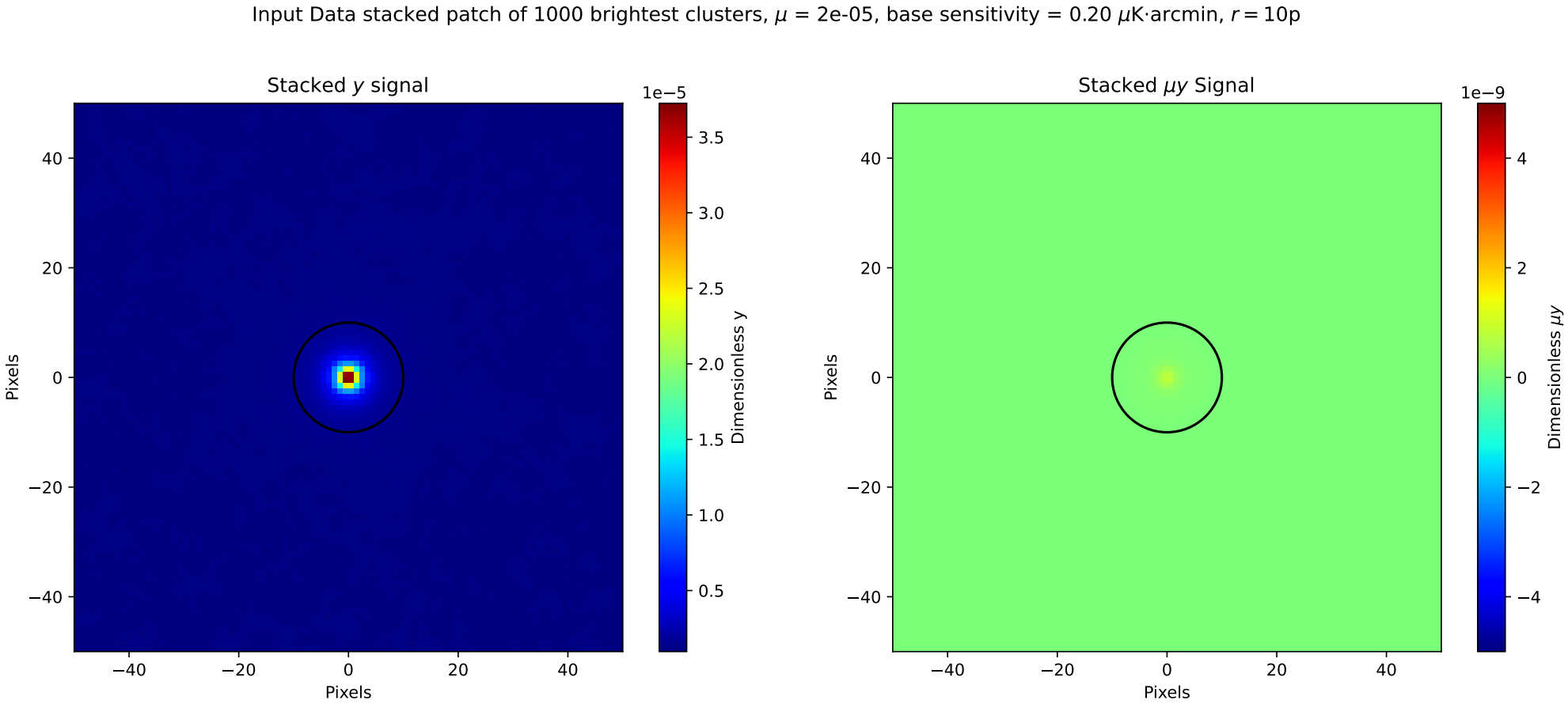


Input stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3



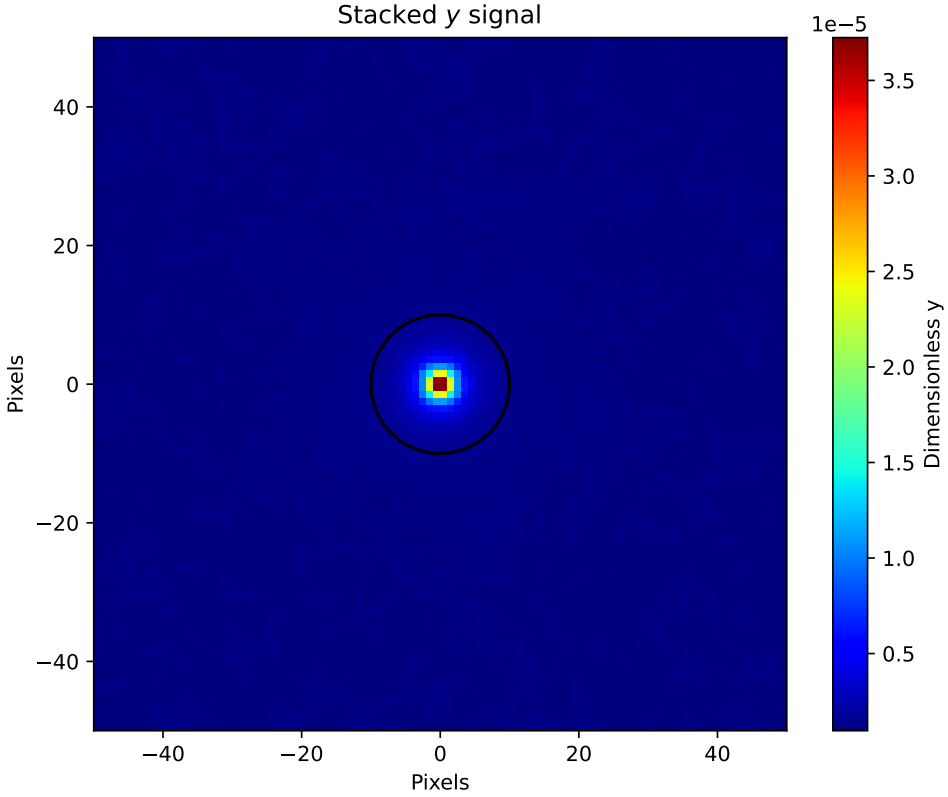
ILC stacked zoomed maps

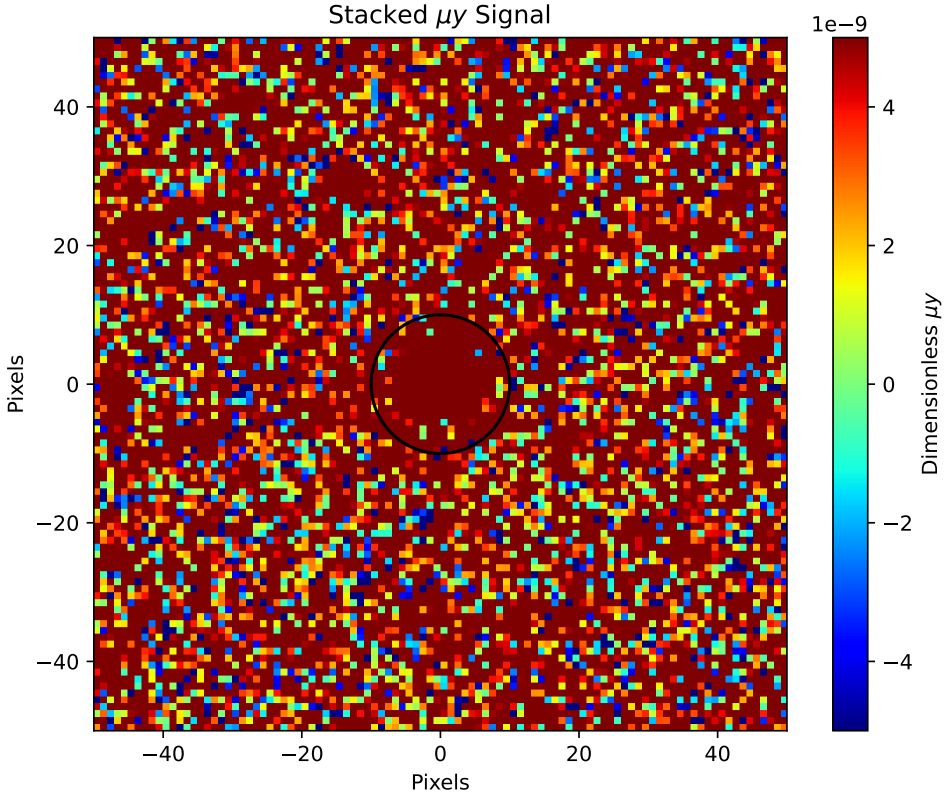
4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-ILC stacked patch of 1000 brightest clusters, μ = 2e-05, base sensitivity = 0.20 μK·arcmin, r = 10p





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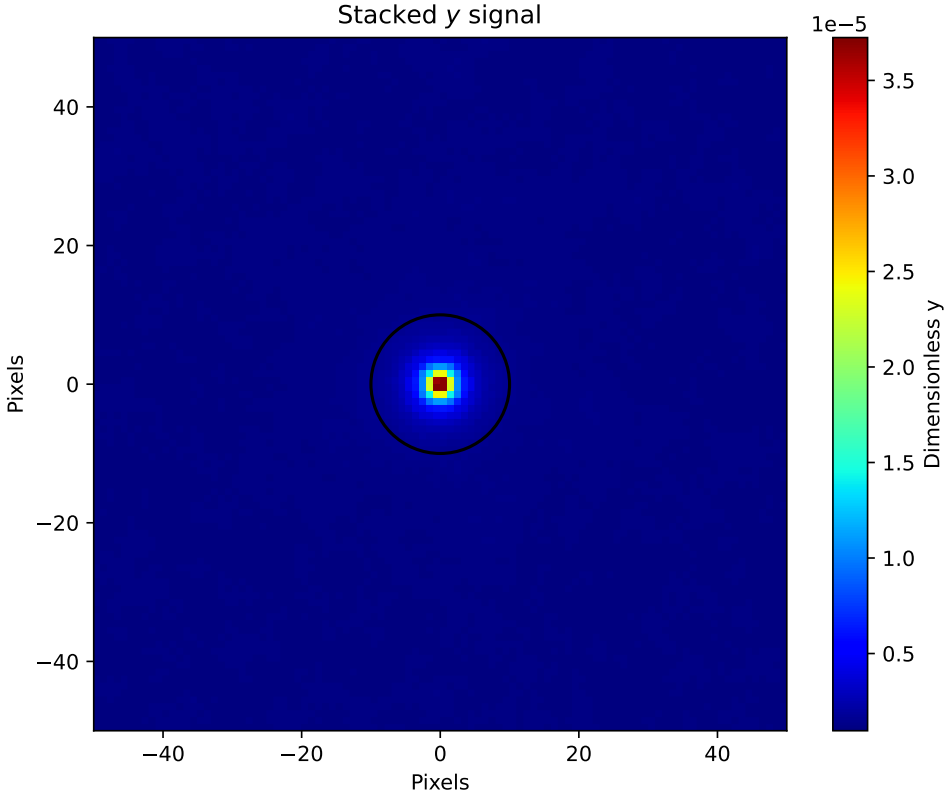
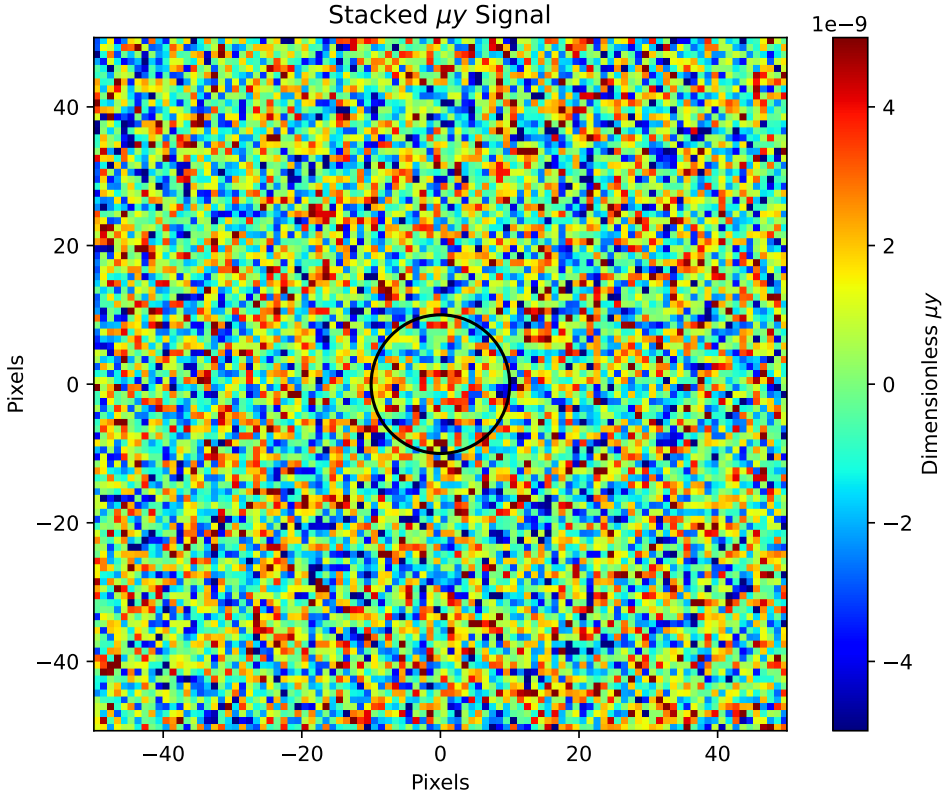


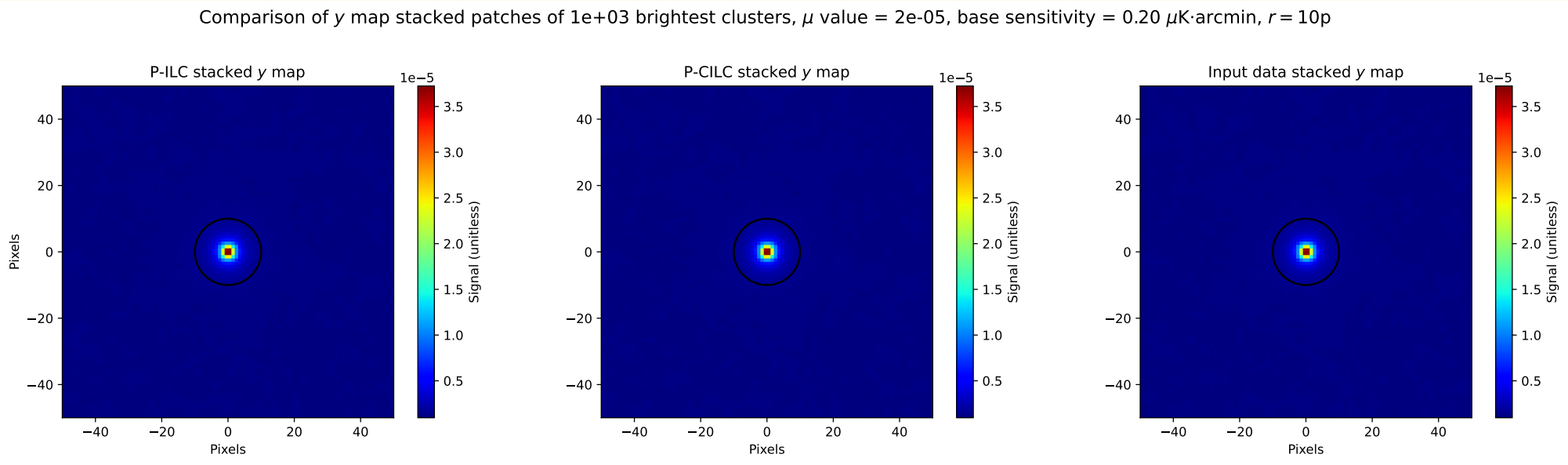


Comparison y stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

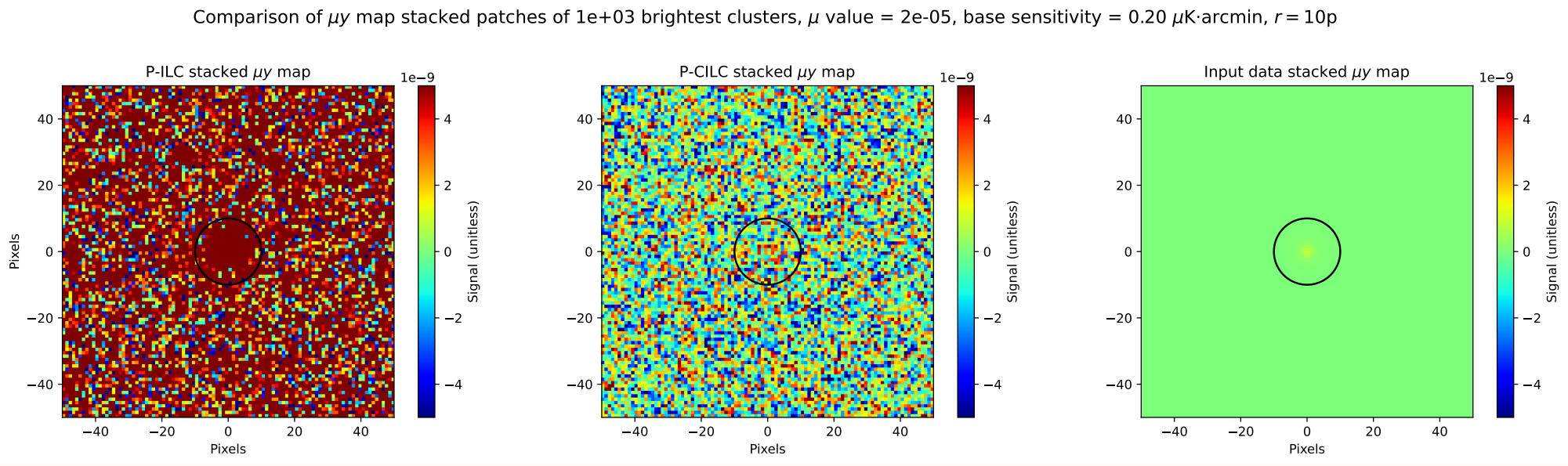


Comparison μy stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

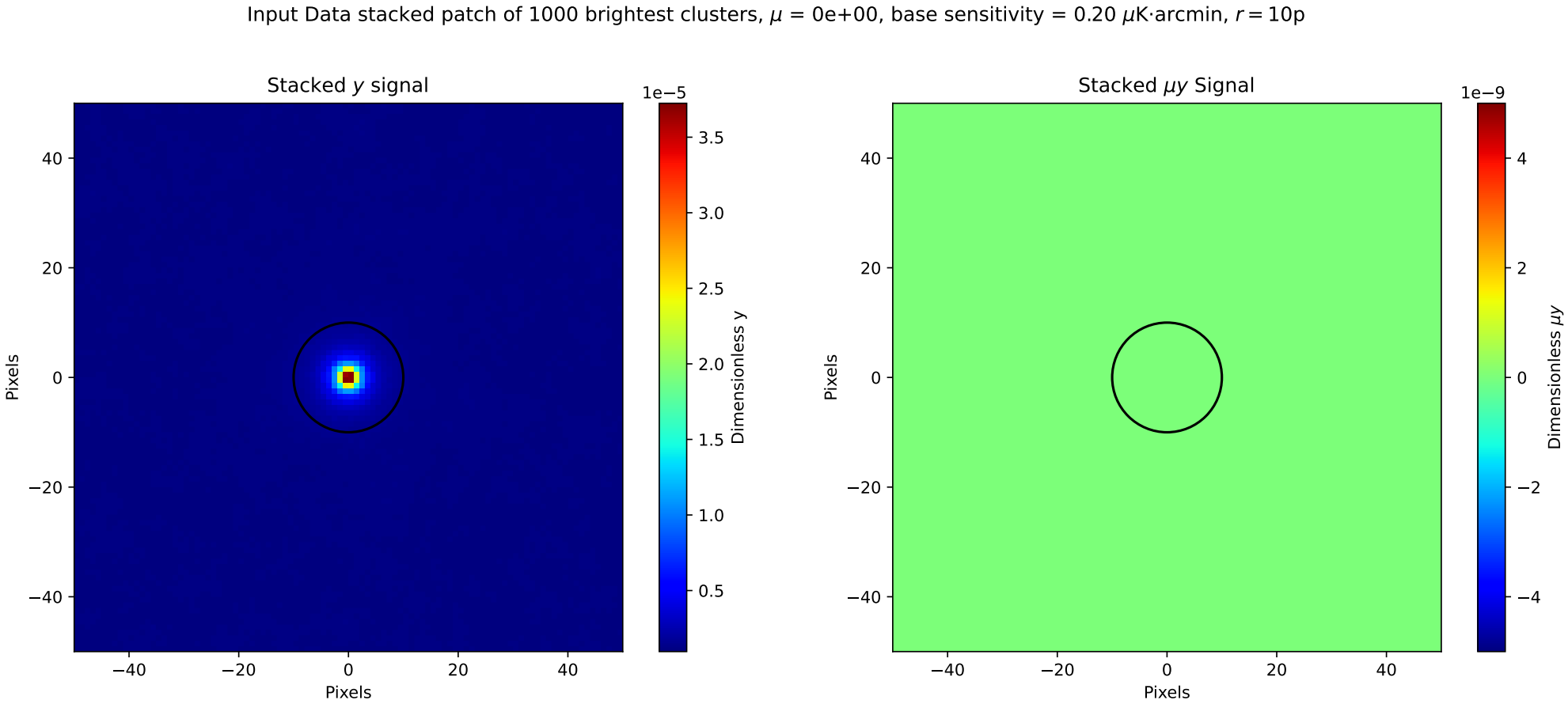


Input stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3



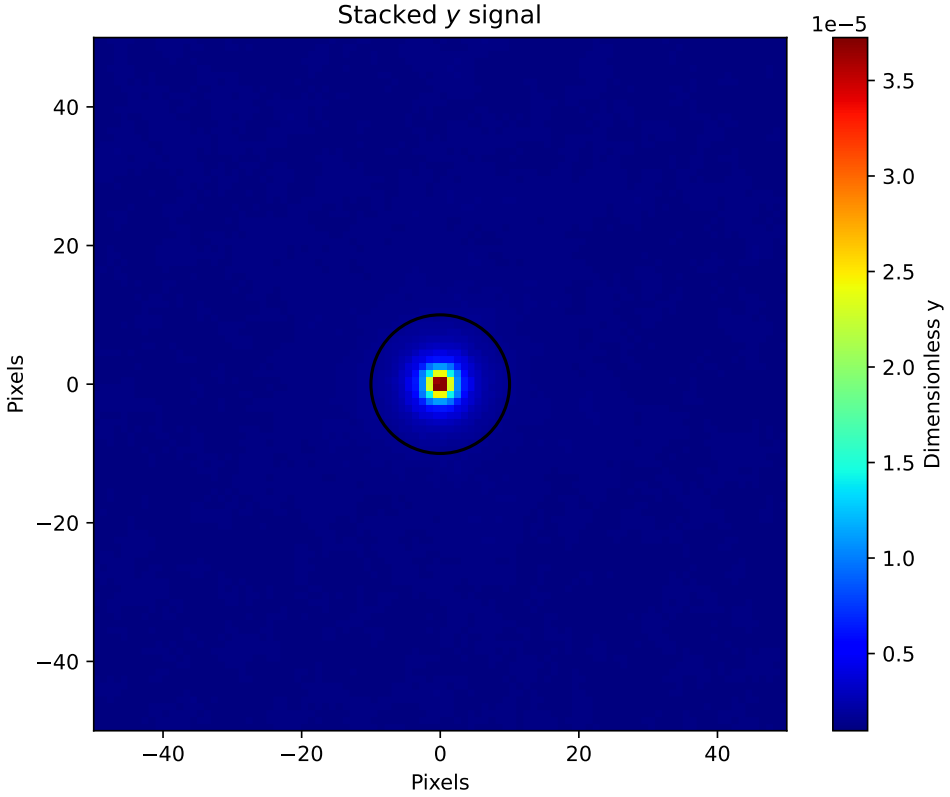
ILC stacked zoomed maps

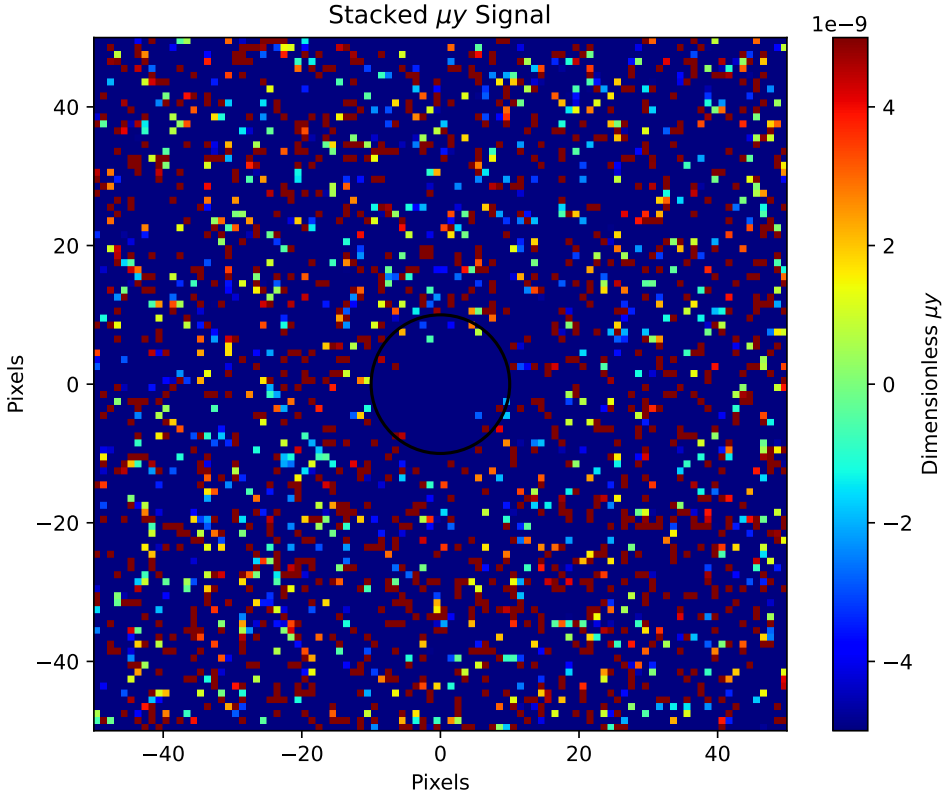
4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-ILC stacked patch of 1000 brightest clusters, μ = 0e+00, base sensitivity = 0.20 μK·arcmin, r = 10p



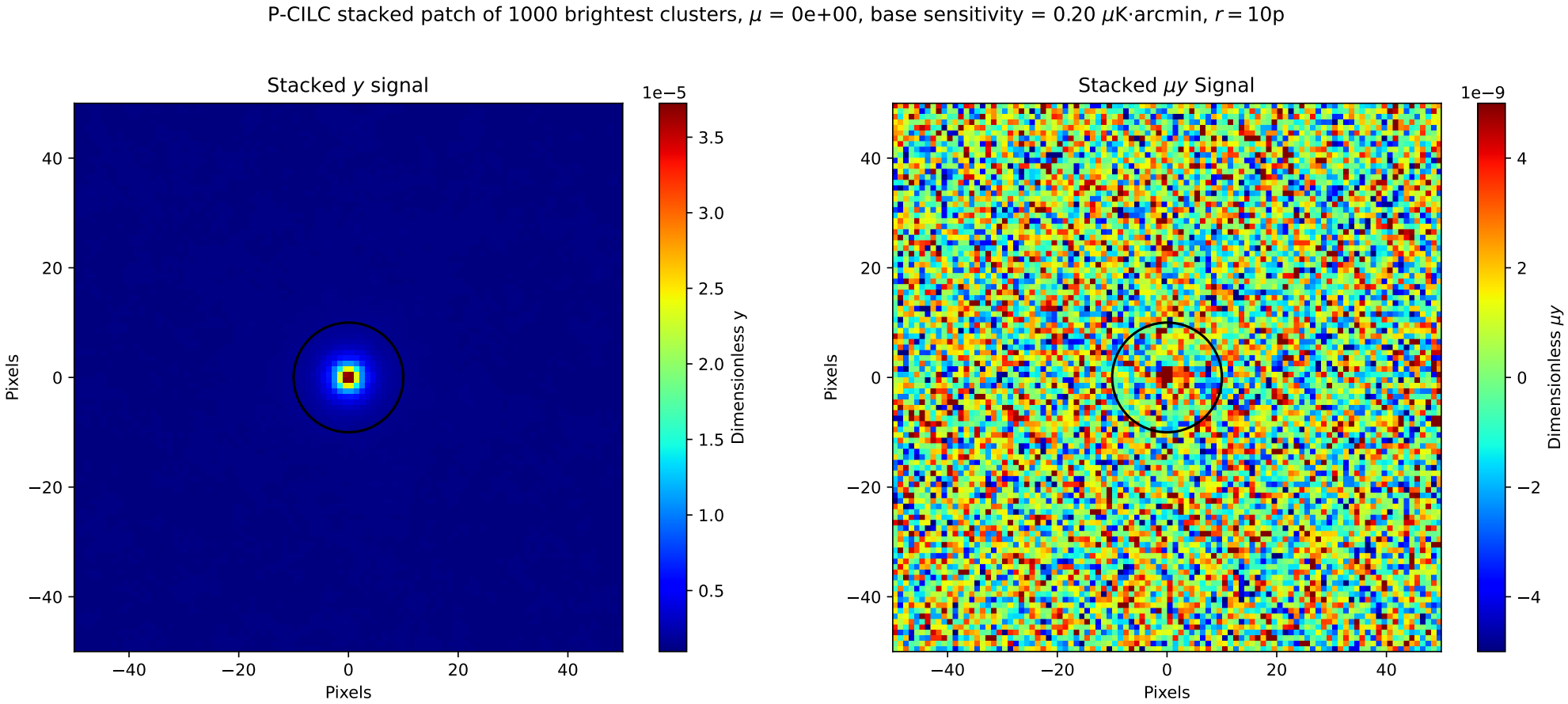


CILC stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

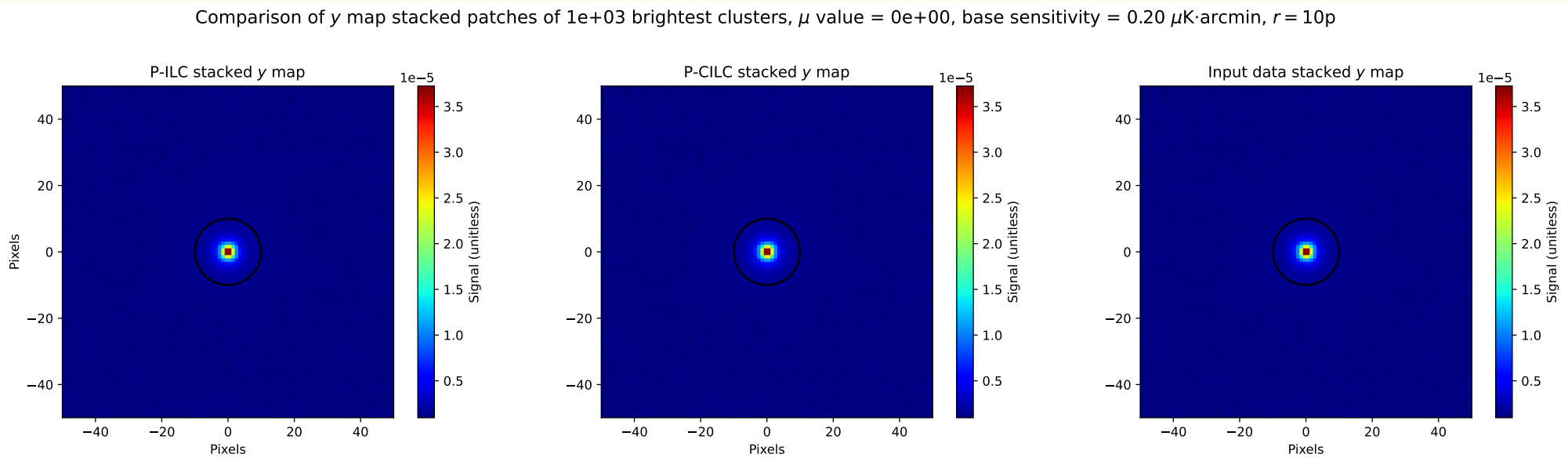


Comparison y stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

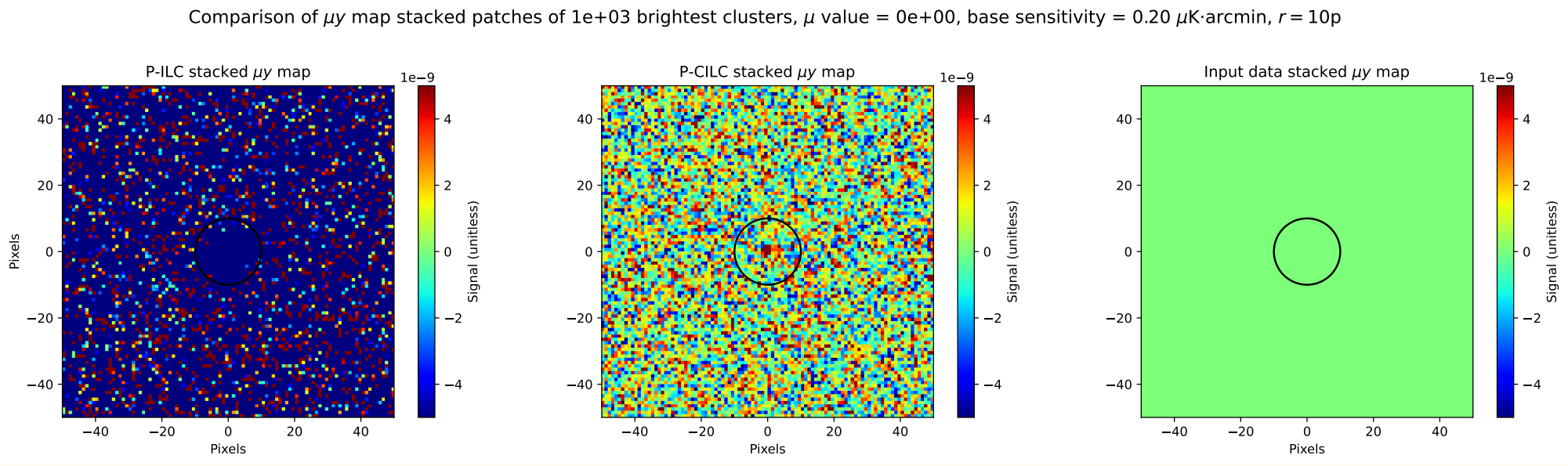


Comparison μy stacked zoomed maps

4 layers: tSZ, μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

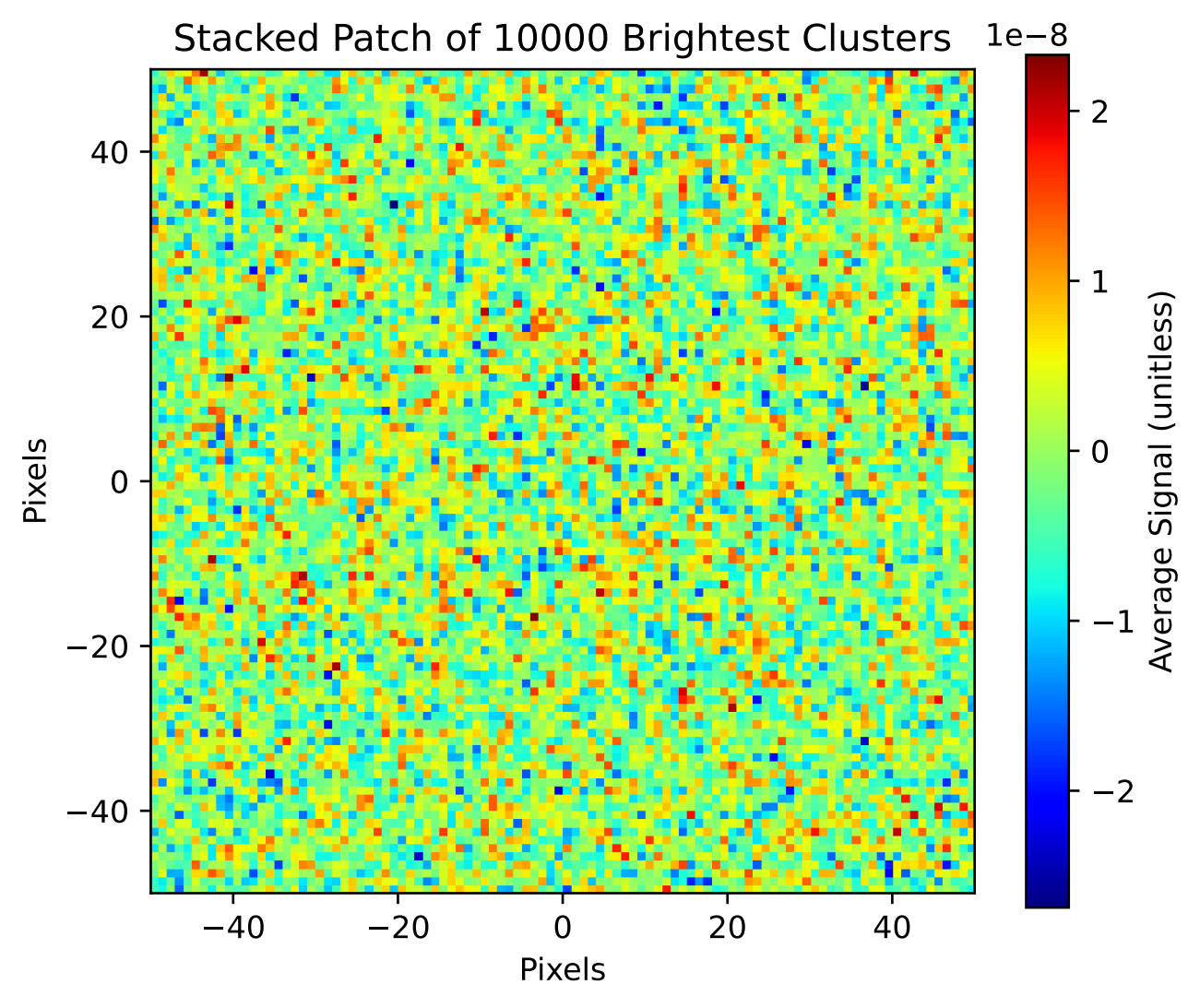
Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3



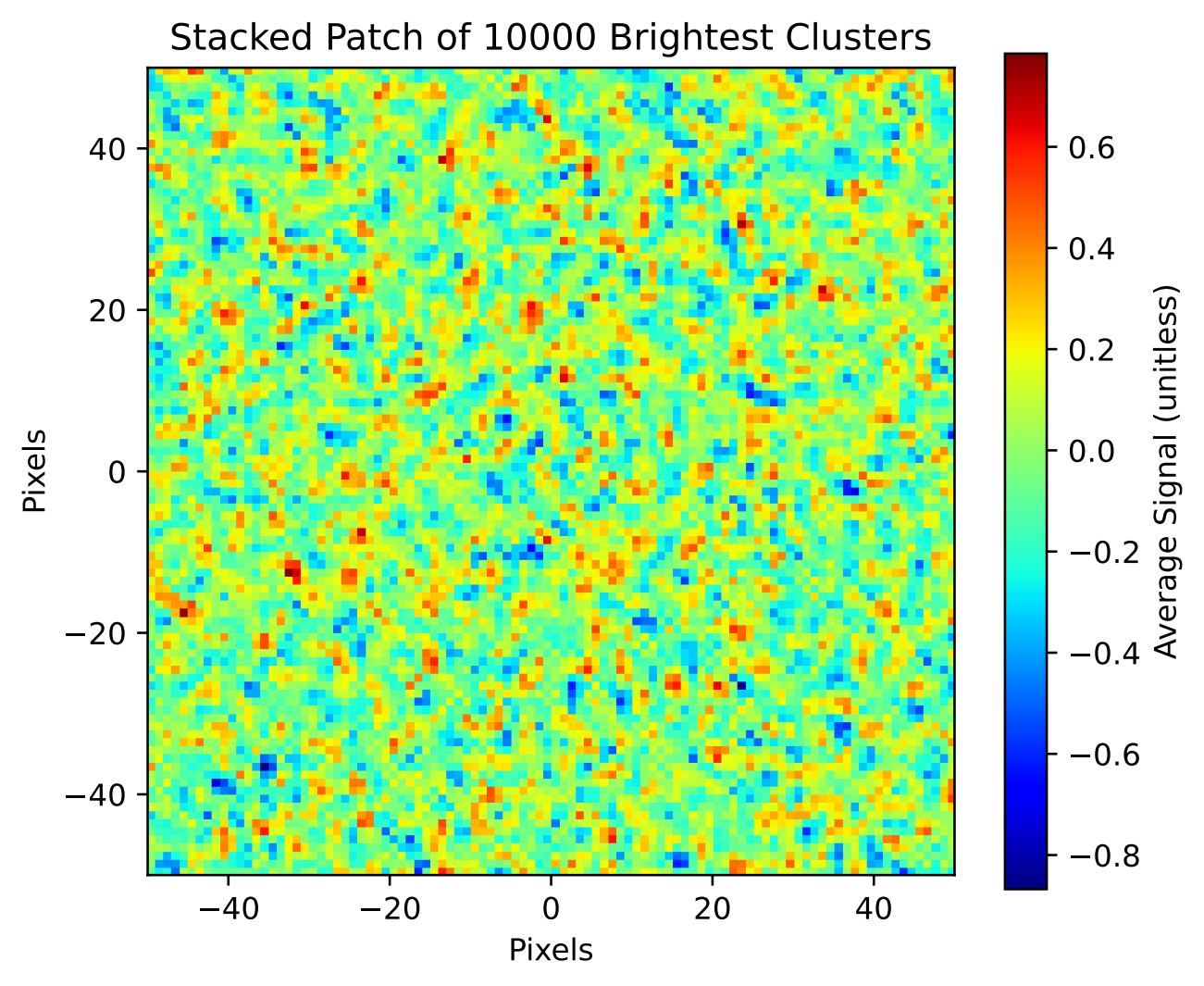
P-ILC stacked zoomed maps (no y input)

3 layers: μ-distortion, CMB and noise ($30\mu K\cdot$arcmin)
$\mu = 2 \times 10^{-8}$, freqs = [30, 40, 90, 150, 220, 270] GHz



H-ILC stacked zoomed maps (no y input)

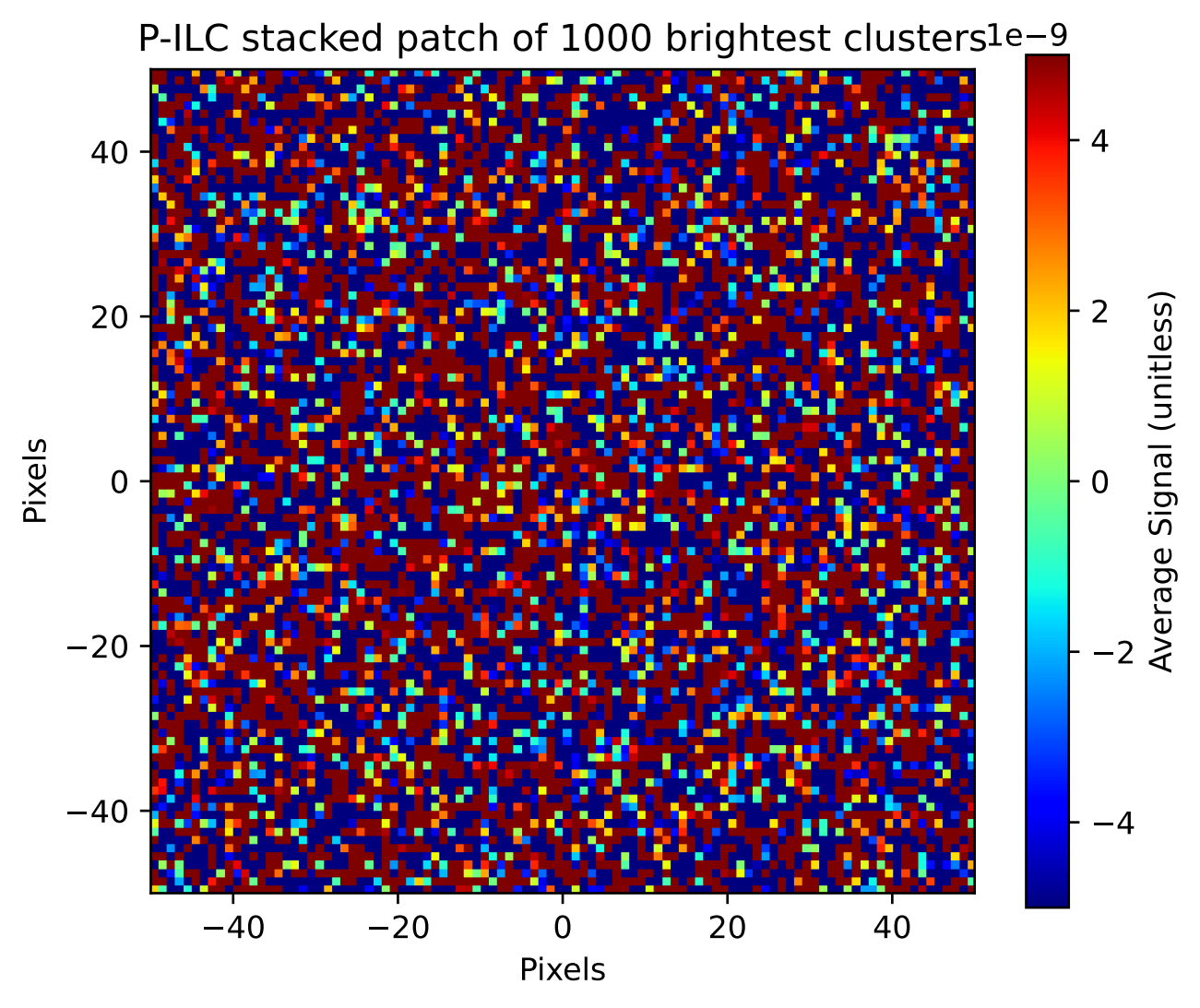
3 layers: μ-distortion, CMB and noise ($30\mu K\cdot$arcmin)
$\mu = 2 \times 10^{-8}$, freqs = [30, 40, 90, 150, 220, 270] GHz



P-ILC stacked zoomed maps (no y input)

3 layers: μ-distortion, CMB and noise ($2\mu K\cdot$arcmin)

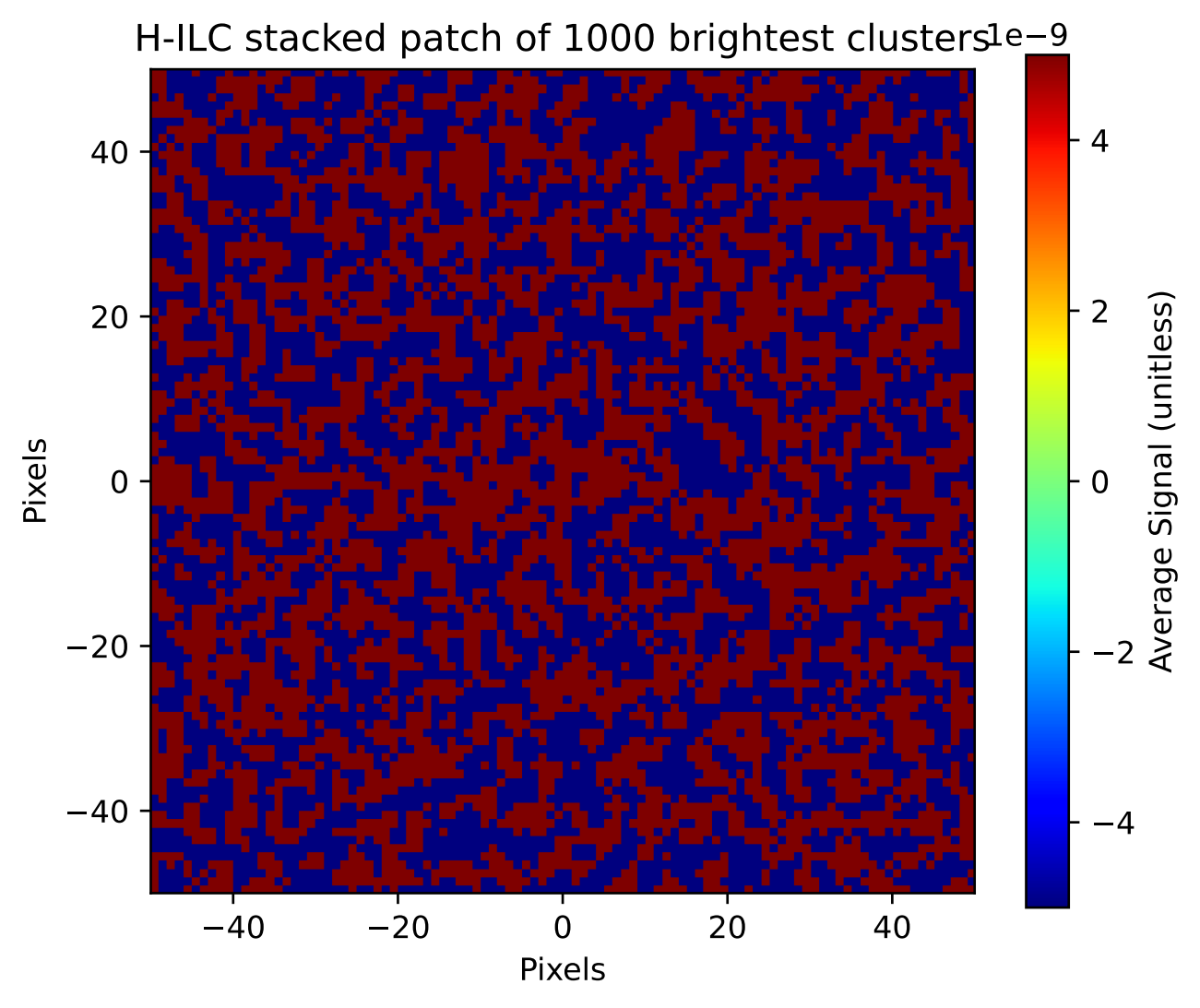
$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



H-ILC stacked zoomed maps (no y input)

3 layers: μ-distortion, CMB and noise ($2\mu K\cdot$arcmin)

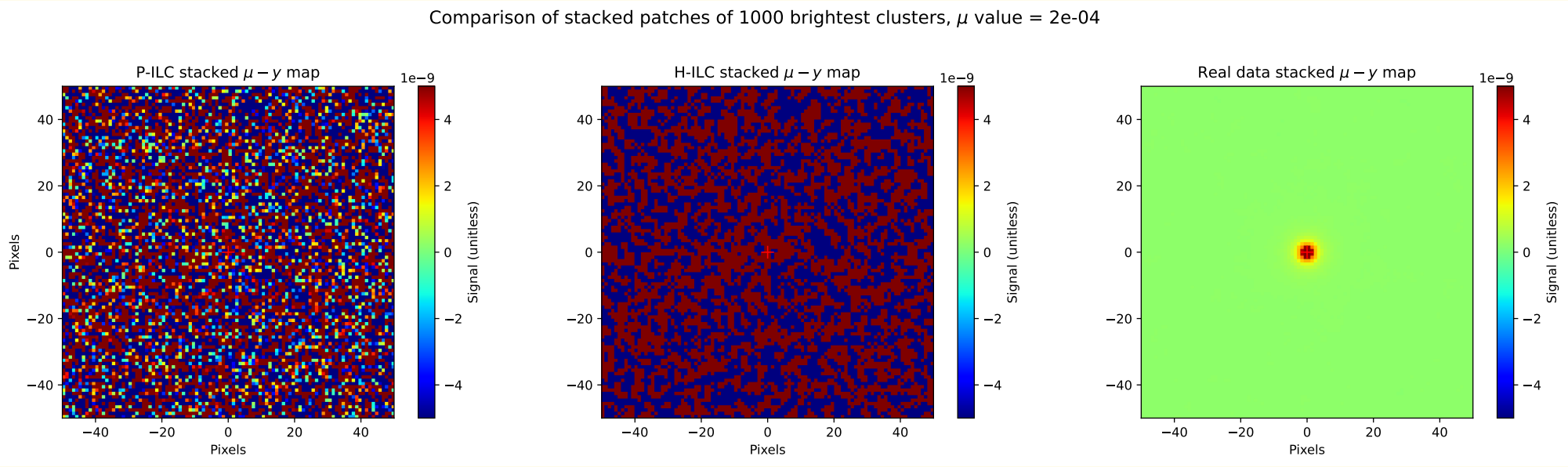
$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



Comparison of stacked zoomed maps (no y input)

3 layers: μ-distortion, CMB and noise ($2\mu K\cdot$arcmin)

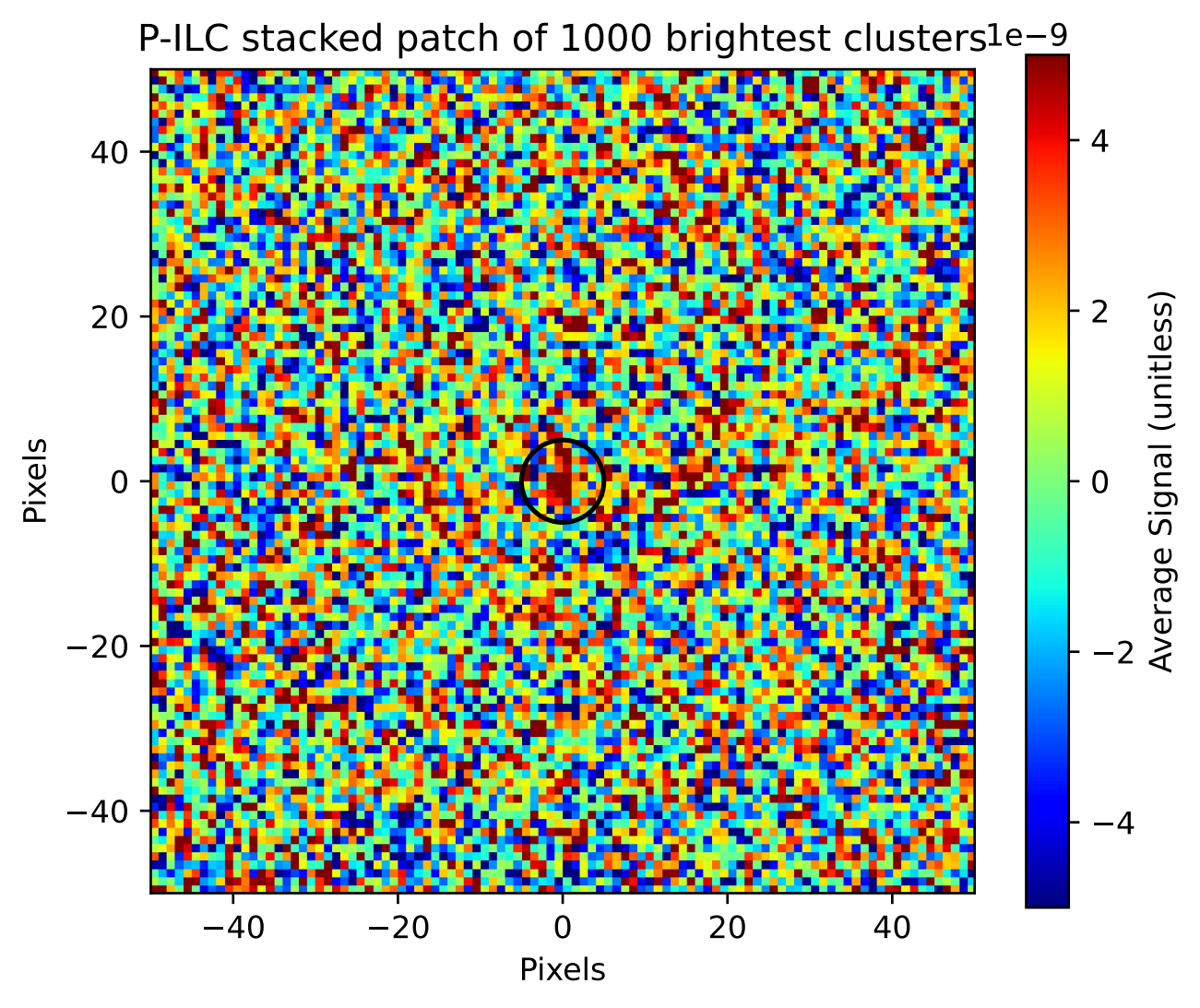
$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



P-ILC stacked zoomed maps (no y input)

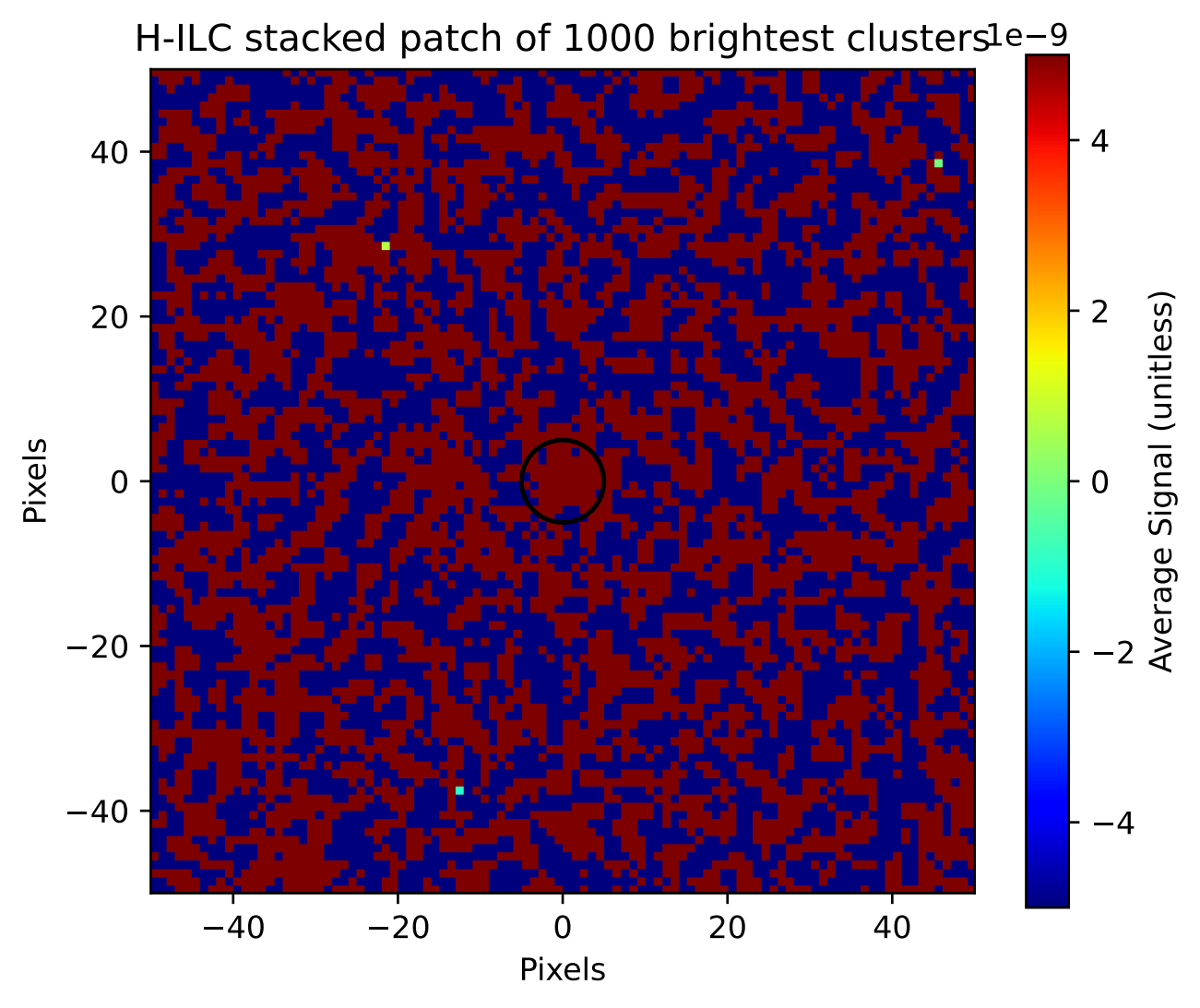
3 layers: μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



H-ILC stacked zoomed maps (no y input)

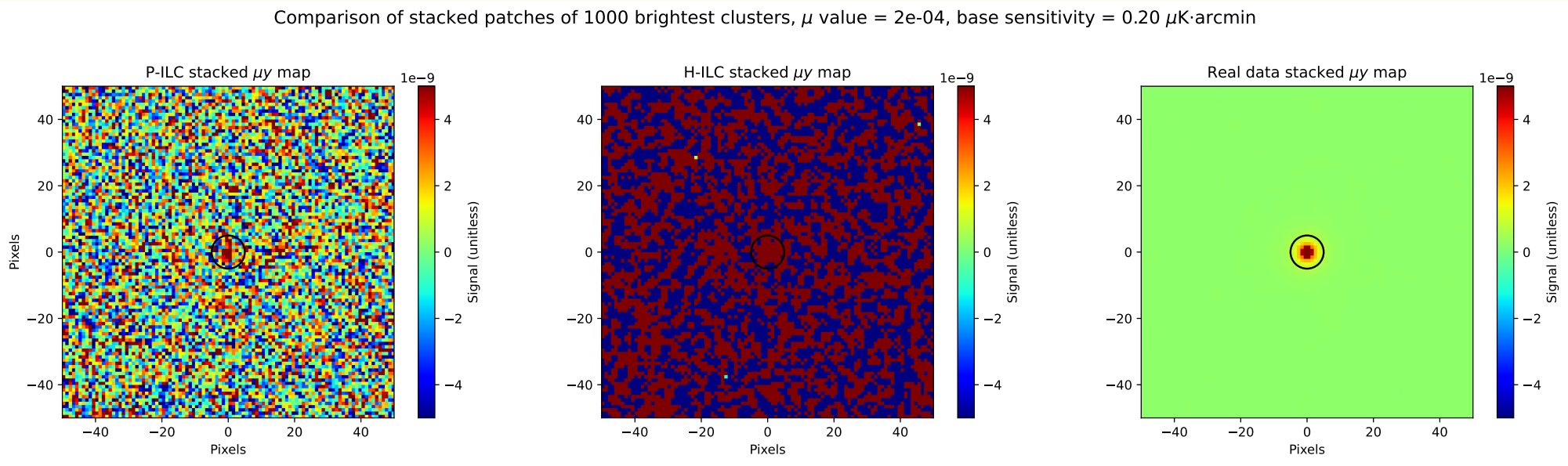
3 layers: μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)
$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



Comparison of stacked zoomed maps (no y input)

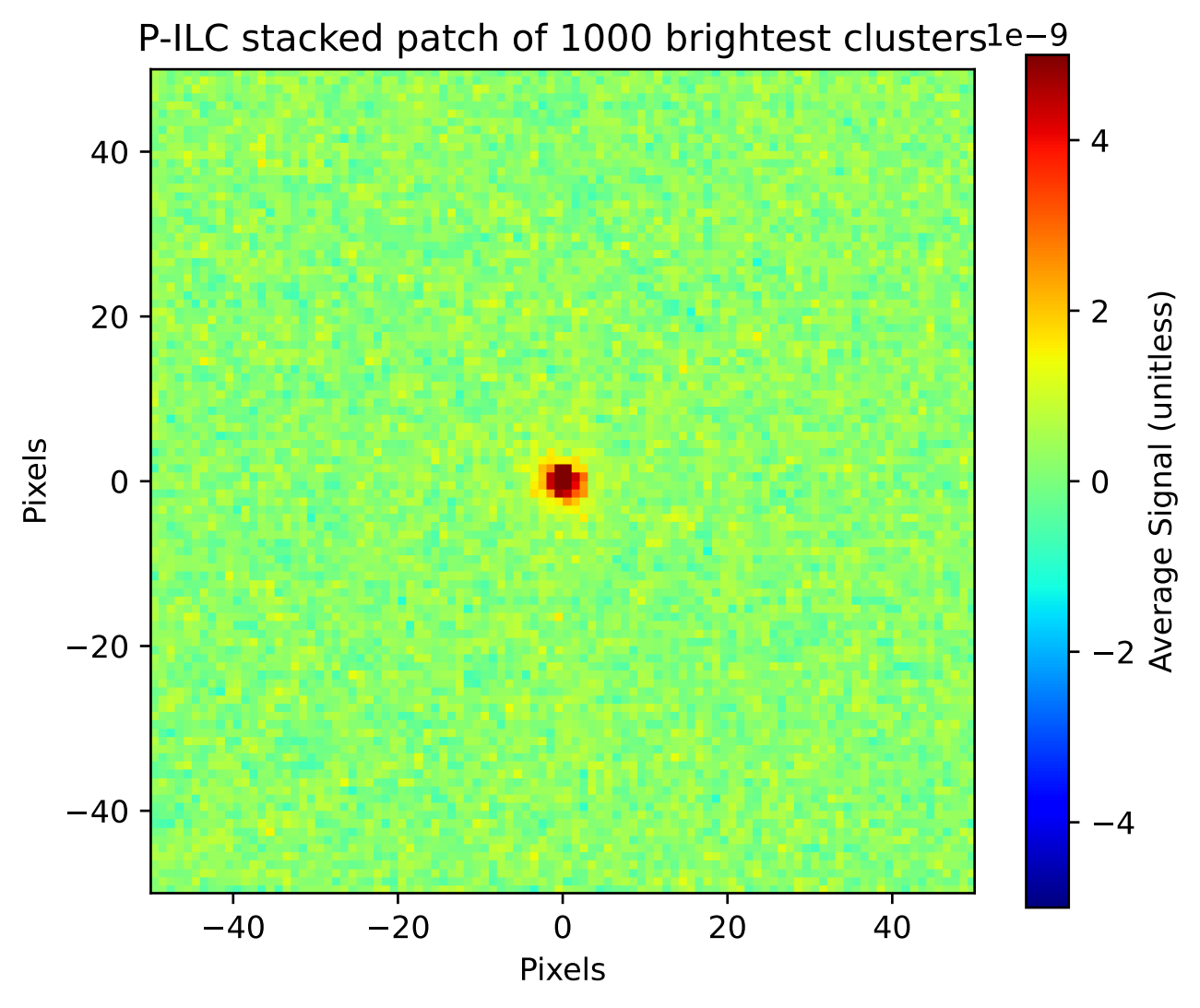
3 layers: μ-distortion, CMB and noise ($0.2\mu K\cdot$arcmin)

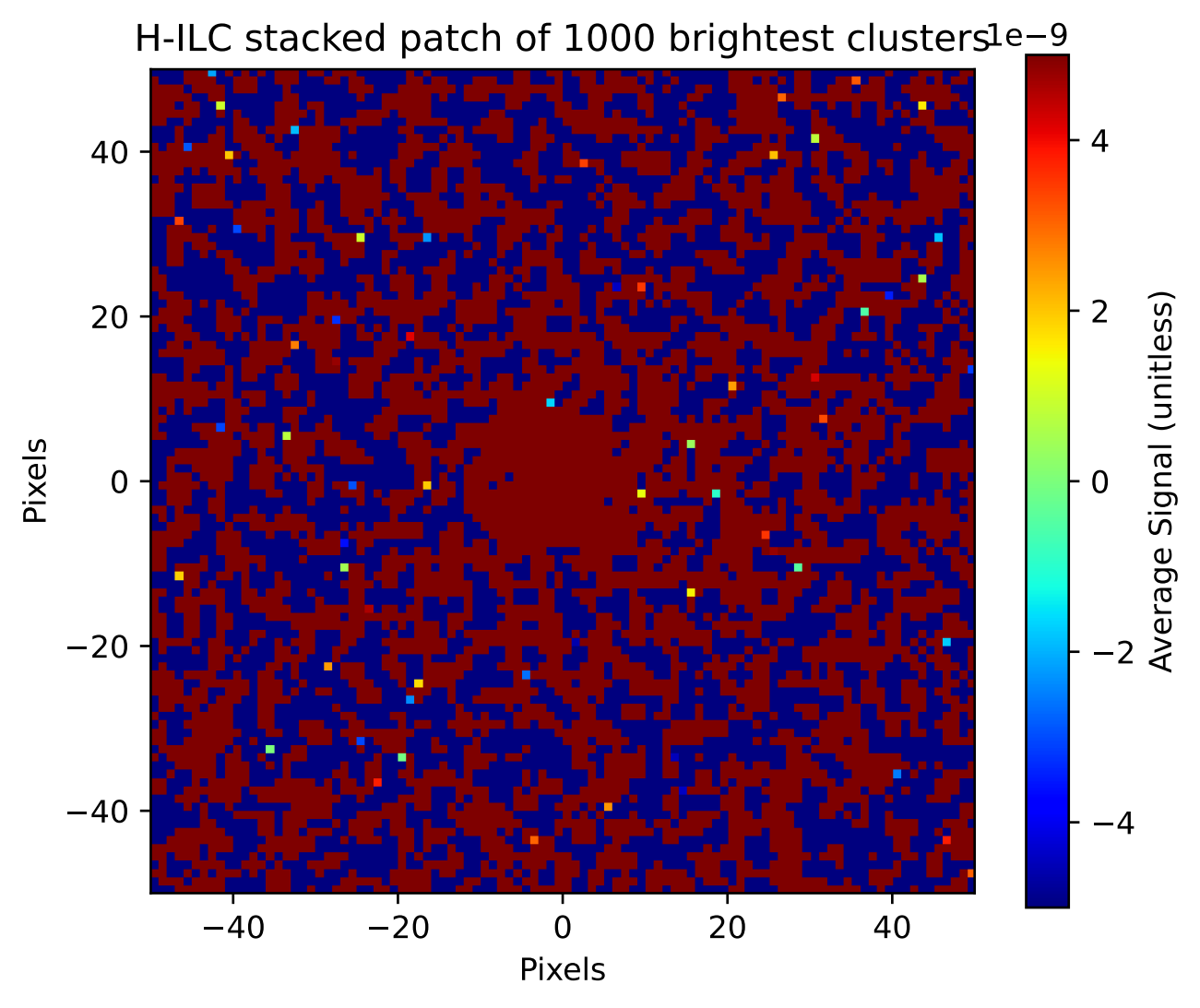
$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



P-ILC stacked zoomed maps (no y input)

3 layers: μ-distortion, CMB and noise ($0.02\mu K\cdot$arcmin)
$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



H-ILC stacked zoomed maps (no y input)

3 layers: μ-distortion, CMB and noise ($0.02\mu K\cdot$arcmin)
$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz